



UNIVERSITY OF WESTERN AUSTRALIA

MASTER OF PHYSICS (COMPUTATIONAL PHYSICS)

**Quantum Circuits for Solving Differential Equations via
Physics-Informed Effective Hamiltonians and GQSP
Quantum Imaginary Time Evolution**

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27 September 2026

Declaration

This is to certify that:

1. This thesis comprises of my original work.
2. Due acknowledgment has been made in the text to all other materials used.
3. This thesis consists of less than 60 pages inclusive of tables and figures but exclusive of references and appendices.

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Acknowledgments

Thank you to my supervisor, Jingbo Wang, for her guidance and support throughout my research, particularly for her continual encouragement and belief in me.

I would also like to thank my family and friends for their support and encouragement throughout my studies.

Summary of Student Achievement

Abstract

Brief summary.....

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Notation and Conventions

A quick reference for symbols used across the thesis; each is properly introduced at its first use in the text below.

Discretisation

Symbol	Meaning	Introduced
n	Qubits per Chebyshev axis (the resolution).	Section 3.2
$N = 2^n$	Truncation dimension of one Chebyshev axis.	Section 3.2
k	Order of an ordinary differential equation.	Section 3.3
p	Degree of a polynomial nonlinearity.	Section 7.3
n_f	Qubits in the field (block) register of the descriptor form.	Section 6.3
$\dim w$	Padded descriptor state dimension, $2^{\lceil \log_2(k+1) \rceil} N$.	Appendix G.2

The Chebyshev model

Symbol	Meaning	Introduced
T_ℓ	Chebyshev polynomial of the first kind, degree ℓ .	Section 3.2
$ \tau(x)\rangle$	The raw Chebyshev feature state.	Section 3.2
ψ, η, η_e	Normalised coefficient state; its scale; $\eta_e = \ b\ ^2$.	Section 3.2
$\hat{f}^{(j)}$	Coefficients of $f^{(j)}$: one block of the descriptor state.	Section 6.3
w	The descriptor state $(\hat{f}^{(0)}; \dots; \hat{f}^{(k)})$.	Section 6.3

Operators and constraints

Symbol	Meaning	Introduced
$\mathbb{G}$	Differentiation matrix, $f \mapsto f'$, in coefficient space: dense, strictly upper triangular, $\ \mathbb{G}\ _2 = \Theta(N^2)$ (Wu et al. print it as $\mathbb{G}_n^\top$).	Section 3.3
$\tilde{\mathbb{B}}, \tilde{\mathbb{D}}$	Banded factors of the differentiation matrix, $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$.	Section 6.2
$\mathbb{M}_{x^p}, \mathbb{M}_a$	Multiplication by x^p (the exact lift, $2N \times N$) and by a smooth $a(x)$.	Section 3.3
$\mathfrak{m}_1, \mathfrak{m}_x$	Product-to-sum folds.	Section 3.5
$\mathbb{B}(x), \mathbb{D}^{(0)}(x_s)$	Rank-one zero (invariant) and collapse (regular) constraint rows.	Section 3.3
x_z, x_m, x_s	Constraint points: a zero, an extremum, a source datum.	Section 3.3
$R_0, U_{\text{odd}}, \Lambda$	Factors of the structured exact encoding of $\mathbb{G}$.	Section 5.2
$\mathbb{U}$	The Chebyshev–Gauss transform used by the nodal fold.	Section 7.4
$\ \mathbb{G}\ _S$	The paper’s norm $\max(\ \mathbb{G}\mathbb{G}^\top\ _1, \ \mathbb{G}^\top\mathbb{G}\ _1)$.	Section 3.4

Residuals and Hamiltonians

Symbol	Meaning	Introduced
$A_{\text{std}}, A_{\text{desc}}$	Residual of the standard form; of the descriptor form.	Section 3.3
$H_{\text{std}}, H_{\text{desc}}$	Effective Hamiltonian of each form; bare H only in regime-independent statements.	Section 3.3
$R = [R_1; \dots; R_L]$	Stacked residual, with $H = R^\dagger R$.	Section 4.4
$H_\otimes$	The doubled-space Hamiltonian.	Section 7.2

Block encoding, filtering and state preparation

Symbol	Meaning	Introduced
$\alpha, \alpha_R, \alpha_H = \alpha_R^2$	Subnormalisation: generic, of R , of H .	Section 2.4
V, W	Stacked-residual reflection; qubitised walk.	Section 4.4
$H' = 2H/\alpha_R^2 - I$	The normalised Hamiltonian that V block-encodes.	Section 4.4
Δ	Spectral gap $\lambda_1 - \lambda_0$.	Section 4.5
β	Imaginary time.	Section 4.5
d, q_k	GQSP degree; filter coefficients.	Section 4.5
ε	Target infidelity (10^{-6}).	Section 4.5
φ_0, γ	Initial state; its overlap $ \langle \psi_{\text{ref}} \varphi_0 \rangle $ with the reference.	Section 4.5
p_G	GQSP success probability.	Section 4.5
ζ	Block-rescaling constant.	Section 6.5
$\epsilon_{\mathbb{G}}, \mu$	Entry error and per-bit coefficient of the published encoding of $\mathbb{G}$.	Appendix D.2

Nonlinearity

Symbol	Meaning	Introduced
K	Carleman truncation order.	Section 7.3
M	Dimension of the base state u .	Section 7.3
ρ	Nonlinearity-to-dissipation ratio; the Carleman precondition is $\rho < 1$.	Section 7.3
$F_1, F_2, \mathbb{A}_K$	Linear and quadratic terms; the lifted Carleman generator.	Section 7.3

1 Introduction

1.1 Differential Equations as a Target for Quantum Computation

1.2 Physics-Informed Hamiltonians and Where Their Cost Lies

1.3 Contributions and Scope

1.4 Outline

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2 Background and Literature Review

2.1 Quantum Computation Preliminaries

In classical computing, the fundamental unit of information is the bit, which can take one of two values: 0 or 1.

The fundamental unit of information in quantum computing is the qubit, which can exist in a superposition of states, allowing it to encode both 0 and 1 simultaneously. This property enables quantum computers to perform certain computations more efficiently than classical computers.

Definition 1 (State Space): The set of all quantum states that a quantum system can occupy is a complex vector space with inner product structure known as a Hilbert space (denoted $\mathcal{H}$). A finite dimensional Hilbert space is isomorphic to $\mathbb{C}^N$ ($\mathcal{H} \cong \mathbb{C}^N$) for some $N \in \mathbb{N}$, where N is the dimension of the Hilbert space. A quantum state is a unit vector in this Hilbert space, represented in dirac notation as $|\psi\rangle \in \mathcal{H}$, with $\langle\psi|\psi\rangle = 1$ (normalisation condition). A quantum state $|\psi\rangle$ can be expressed as a linear combination of basis states $\{|i\rangle\}_{i=0}^{N-1}$, where $|i\rangle$ is the i^{th} basis state of the Hilbert space.

$$|\psi\rangle = \sum_{i=0}^{N-1} \psi_i |i\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{N-1} \end{pmatrix} \quad (2.1)$$

Its hermitian conjugate (or adjoint) is denoted as $\langle\psi| = |\psi\rangle^\dagger$. Note that for some $e^{i\theta}$ for $\theta \in [0, 2\pi)$, the states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically indistinguishable, and thus correspond to the same physical quantum state. This is known as a global phase factor.

Definition 2 (The Qubit): A quantum bit (qubit) is a two-state quantum system, characterised by $|\psi\rangle \in \mathbb{C}^2$.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (2.2)$$

where $\alpha, \beta \in \mathbb{C}$, $|0\rangle$ and $|1\rangle$ are the computational basis states, and $|\alpha|^2, |\beta|^2$ are the probabilities of measuring the qubit in the $|0\rangle$ and $|1\rangle$ states, respectively. The normalisation condition requires that $|\alpha|^2 + |\beta|^2 = 1$.

Definition 3 (Composite Systems): The state space of a composite quantum system is the tensor product of the state spaces of its subsystems. Consider two quantum states $|\psi\rangle_A \in \mathcal{H}_A$

and $|\psi\rangle_B \in \mathcal{H}_B$, the composite state $|\psi\rangle_{AB}$ is given by:

$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (2.3)$$

where $\otimes$ denotes the tensor product and is often hidden by convention. The dimension of the composite state space is the product of the dimensions of the individual state spaces. If a composite system cannot be written as a tensor product of its subsystems, it is said to be entangled. Entanglement is a unique feature of quantum mechanics that has no classical counterpart and plays a crucial role in quantum information processing.

We often consider the state of a composite system of n qubits, which is represented in a 2^n -dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^{2^n}$.

2.1.1 Quantum Gates and Circuits

Definition 4 (Quantum Algorithm): A quantum algorithm is a sequence of operations that manipulates qubits to perform a specific computation.

The goal of any quantum algorithm is to manipulate quantum states in a controlled manner to achieve a desired outcome. This is accomplished through the application of quantum gates, which are unitary operations that transform quantum states. Quantum gates can be combined to form quantum circuits, which represent the sequence of operations performed on qubits. Each quantum gate can be implemented using physical operations on hardware, and the design of efficient quantum algorithms is a central challenge in quantum computing research.

Definition 5 (Unitary Evolution): The evolution of a closed quantum system between two states is described by unitary transformations $U \in \mathbb{C}^{2^n \times 2^n}$, where n is the number of qubits in the system.

$$|\psi\rangle_{final} = U |\psi\rangle_{initial}, \quad U^\dagger U = U U^\dagger = I \quad (2.4)$$

where $U^\dagger$ is the hermitian conjugate of U and I is the identity operator. Unitary transformations preserve the inner product and thus the norm of quantum states, ensuring that states have a physical interpretation as probability amplitudes.

These unitary transformations are known as quantum gates, and each gate serves a specific purpose in manipulating quantum states.

The quantum circuit diagram offers a succinct and intuitive representation of quantum algorithms, analogous to Feynman diagrams in quantum field theory.

Time flows left to right, and each wire (horizontal line) represents a qubit or a quantum register. Gates are placed sequentially on the wires indicating operations applied to qubits. An example is the Hadamard gate, which creates superposition, and the CNOT gate, which entangles qubits. A full list of quantum gates and their corresponding circuit symbols can be found in .

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2.1.2 Complexity of Quantum Algorithms

Definition 6 (Gate Complexity): The total number of quantum gates used in a quantum circuit.

Definition 7 (Circuit Depth): The maximum number of gates applied sequentially on any qubit in a quantum circuit.

Definition 8 (Ancillary Qubits): Additional qubits introduced into a quantum circuit beyond the primary input qubits, typically initialised to a baseline state (such as $|0\rangle$) to assist with intermediate calculations, temporary storage, or entanglement generation.

Definition 9 (Big- $\mathcal{O}$ Notation): Big- $\mathcal{O}$ notation is a mathematical notation used to describe the asymptotic behaviour of functions.

$$\mathcal{O}(f(n)) = \{g(n) : \exists c > 0, n_0 \in \mathbb{N} \text{ such that } 0 \leq g(n) \leq cf(n) \text{ for all } n \geq n_0\} \quad (2.5)$$

In the context of quantum algorithms, it is used to express the scaling of gate complexity, circuit depth, and space requirements with respect to the number of qubits n .

For example, if a quantum algorithm has a gate complexity of $\mathcal{O}(n^2)$, it means that the number of gates required grows quadratically with the number of qubits.

A quantum circuit is said to be **efficient** if its circuit depth, gate complexity, and the number of ancillary qubits all scale polynomially with the number of qubits n .

2.1.3 Noisy Intermediate-Scale Quantum (NISQ) Era

Write this section. Use Josh's as a guide.

Table 1: Quantum approaches to differential equations: what each assumes, what it costs, and where it fails.

Approach	Assumptions	Query cost	Output	Failure mode
Linear systems (HHL family)				
Linear combination of Hamiltonian sim- ulations				
Carleman linearisa- tion				
Variational				
Physics-informed Hamiltonian [1]				
This work				

2.2 Quantum Algorithms for Differential Equations

2.2.1 Linear-Systems and Linear-Combination Routes

2.2.2 Nonlinear Equations: Carleman and Related Linearisations

2.2.3 Variational and Ground-State Routes

2.2.4 Critical Assessment

2.3 Spectral Methods and Banded Differentiation

2.3.1 Chebyshev Series, Differentiation and Galerkin Truncation

2.3.2 The Ultraspherical Spectral Method

2.3.3 Fourier–Galerkin Discretisation of Periodic Flows

2.4 Block Encoding, Linear Combinations of Unitaries and Qubitisation

2.5 Quantum Signal Processing and Its Generalisation

2.6 Ground-State Preparation by Filtering

2.7 Readout and Amplitude Amplification

2.8 Synthesis

3 The Physics-Informed Effective Hamiltonian Method

3.1 Overview

Wu et al. recast solving a differential equation as three linked moves [1]: represent a candidate solution as a quantum state carrying its Chebyshev coefficients; write the differential equation and every boundary or initial condition as linear operators that must annihilate that state; and assemble a Hamiltonian from those operators whose ground state is exactly the state satisfying all of them at once.

3.2 The Latent Chebyshev Model

The Chebyshev feature map. A solution f is approximated by a degree- $(N-1)$ truncated Chebyshev series, $N = 2^n$, carried by an n -qubit state

$$|\tau(x)\rangle_n = \frac{1}{2^{n/2}} T_0(x) |0\rangle^{\otimes n} + \frac{1}{2^{(n-1)/2}} \sum_{k=1}^{N-1} T_k(x) |k\rangle, \quad (3.1)$$

so that a coefficient vector $|\psi\rangle_n \in \mathbb{C}^N$ (unit norm, $\langle\psi|\psi\rangle = 1$) represents the model

$$f_Q(x) = \sqrt{\eta} \langle\tau(x)|\psi\rangle_n \quad (3.2)$$

for some positive scale η . Separating the *direction* $|\psi\rangle$ from the *scale* $\sqrt{\eta}$ this way matters because everything from here to the ground-state eigenvalue problem below depends only on $|\psi\rangle$: η cancels out of every constraint $A|\psi\rangle = 0$ and out of the Hamiltonian built from such constraints, so it plays no role until Equation (3.9) fixes it, once, from a single known data point, entirely after the ground state has been found. Every equation from here on is written for $|\psi\rangle$ alone, with η suppressed accordingly.

3.3 The Residual, the Hamiltonian and Its Constraints

The derivative operator. Differentiating a Chebyshev series produces another Chebyshev series whose coefficients are a fixed linear function of the original ones [2]: writing this map in the normalised basis of Equation (3.1) gives an $N \times N$ matrix $\mathbb{G}$ with

$$\mathbb{G}_{k,p} = \begin{cases} 2p, & p > k \geq 1, \text{ } p - k \text{ odd,} \\ \sqrt{2} p, & k = 0, \text{ } p \text{ odd, } p \geq 1, \\ 0, & \text{otherwise,} \end{cases} \quad (3.3)$$

so that $f'_Q(x) = \langle\tau(x)|\mathbb{G}|\psi\rangle_n$, and the m -th derivative follows from the power $\mathbb{G}^m$. Equation (3.3) is dense and upper triangular: row k is non-zero at every column $p > k$ of the right parity, so the lowest-order coefficient can depend on as many as half of the others, and $\mathbb{G}$ carries $\Theta(N^2)$ non-zero entries in total. Appendix B gives this fill-in explicitly at $N = 4$, alongside the banded factors $\tilde{\mathbb{B}}, \tilde{\mathbb{D}}$ that Section 6.2 builds from the same recurrence.

The differential-equation row. For a linear equation of order k with constant coefficients, $\sum_{j=0}^k a_j f^{(j)} = 0$, direct substitution of $\mathbb{G}^j$ for each derivative gives

$$A_{\text{std}} = \sum_{j=0}^k a_j \mathbb{G}^j, \quad (3.4)$$

Wu et al.'s own construction [1]; a second-order equation, for instance, gives $A_{\text{std}} = a_2 \mathbb{G}^2 + a_1 \mathbb{G} + a_0 I$ directly.

A coefficient $a_j(x)$ that varies with x cannot be substituted this way: $a_j(x) f^{(j)}(x)$ is a product of two functions, and the Chebyshev coefficients of a product are not the product of the coefficients. Wu et al. resolve this with a second operator, the lifting operator $\mathbb{M}_{x^p}$, built from the general

Chebyshev product-to-sum identity

$$x^p T_k(x) = 2^{-p} \sum_{i=0}^p \binom{p}{i} T_{|k+p-2i|}(x), \quad (3.5)$$

the p -fold iterate of the pairwise rule $T_j T_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$ that reappears below in the nonlinear extension. Conjugating Equation (3.5) into the normalised bases of Equation (3.1) gives an operator $\mathbb{M}_{x^p}$ satisfying

$$x^p \langle \tau(x) | _n = \langle \tau(x) | _{n+1} \mathbb{M}_{x^p} \quad (3.6)$$

(entries: Appendix C). Multiplying a degree- $(N-1)$ model by x^p can raise its degree by as much as p , so $\mathbb{M}_{x^p}$ is exact and rectangular, $2N \times N$: no truncation is needed once the result is represented in the one-qubit-wider register Equation (3.6) already targets. A general coefficient, expanded to a finite power series $a_j(x) = \sum_p c_{j,p} x^p$ (a Maclaurin truncation, for a non-polynomial coefficient), lifts by linearity to $\mathbb{M}_{a_j} = \sum_p c_{j,p} \mathbb{M}_{x^p}$, and the differential-equation row for the general, variable-coefficient case follows by composing $\mathbb{M}_{a_j}$ and $\mathbb{G}^j$ term-by-term,

$$A_{\text{std}} = \sum_{j=0}^k \mathbb{M}_{a_j} \mathbb{G}^j, \quad (3.7)$$

with $\mathbb{M}_{a_j} = a_j I$ recovering Equation (3.4) whenever a_j is constant.

$$A_{\text{std}} = \sum_{j=0}^k \mathbb{M}_{a_j} \mathbb{G}^j - \mathbb{M}_r \mathbb{D}^{(0)}(x_s), \quad \mathbb{D}^{(0)}(x_s) = \frac{\mathbb{B}(x_s)}{f(x_s)}. \quad (3.8)$$

Data constraints. Boundary and initial conditions are built from one rank-one point-evaluation generator, $\mathbb{B}_n(x) = \sqrt{2^n} |0\rangle^{\otimes n} \langle \tau(x) | _n$, non-zero only in its first row. A condition with a zero target is *invariant* – unchanged under $|\psi\rangle \rightarrow c|\psi\rangle$ – and enters the Hamiltonian directly as one further linear constraint, $\mathbb{B}|\psi\rangle = 0$, e.g. a Dirichlet condition $f(x_z) = 0$ giving $\mathbb{B}_n(x_z)|\psi\rangle = 0$ and a Neumann condition $f'(x_m) = 0$ giving $\mathbb{B}_n(x_m)\mathbb{G}|\psi\rangle = 0$. A condition with a non-zero target, $f(x_s) = y_s$, is not scale-covariant and instead fixes the amplitude $\sqrt{\eta}$ of Equation (3.2) directly, after the ground state has been found:

$$\sqrt{\eta} = \frac{y_s}{\langle \tau(x_s) | \psi_g \rangle_n}. \quad (3.9)$$

Only one such *regular* constraint is used per solve, and Equation (3.9) is the only place η re-enters the pipeline.

Definition 10 (Physics-informed effective Hamiltonian): Collecting the differential-equation

row A_{std} and every invariant data constraint $\mathbb{B}_i$,

$$H_{\text{std}} = A_{\text{std}}^\top A_{\text{std}} + \sum_i \mathbb{B}_i^\top \mathbb{B}_i, \quad \langle \psi | H_{\text{std}} | \psi \rangle = \|A_{\text{std}} \psi\|^2 + \sum_i \|\mathbb{B}_i \psi\|^2. \quad (3.10)$$

H_{std} is Hermitian and positive semi-definite, so $\langle \psi | H_{\text{std}} | \psi \rangle \geq 0$ with equality exactly when every constraint holds simultaneously. The differential equation problem is thereby an eigenvalue problem, $H_{\text{std}} |\psi_g\rangle = \lambda_{\min} |\psi_g\rangle$, $\lambda_{\min} \approx 0$, whose ground state $|\psi_g\rangle$ is the solution.

3.4 The Published Circuits, Filter and Readout

$$\alpha_{\mathbb{G}}^{\text{pub}} = (2^{n-1} + 2^n) \|\mathbb{G}\|_S, \quad \|\mathbb{G}\|_S = \max(\|\mathbb{G}\mathbb{G}^\top\|_1, \|\mathbb{G}^\top\mathbb{G}\|_1), \quad (3.11)$$

$$\beta_{\text{pub}} \geq \frac{\lambda_{\max}}{\lambda_2(n-1)}. \quad (3.12)$$

Optional schematic of the pipeline as published (Figs 2, 8 and 9 of the paper, QSVT-QITE on $H/\|H\|_F$, overlap readout), drawn to match the thesis's own pipeline figure (Figure 2) so the two can be compared at a glance.

Figure 1: The published pipeline of Wu et al.: residual, block encoding, QSVT imaginary-time filter and interferometric readout.

3.5 The Doubled-Space Extension to Nonlinearity

Wu et al. extend Definition 10 to polynomial nonlinearity by doubling the Hilbert space, $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, and folding a product of two coefficient vectors back into one register via the same product-to-sum identity, $T_j T_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$ [1]. A residual quadratic in $|\psi\rangle$ of this kind annihilates a subspace of the $2N$ -dimensional $|\Psi\rangle$ far larger than the single physical direction: Wu et al. themselves note that the resulting H carries *at least* 2^{2n-1} degenerate zero-energy states, half of the doubled Hilbert space at minimum. No imaginary-time filter can select a direction inside a degenerate zero-energy cluster, so extracting the physical solution from this construction is classical post-selection within a known kernel rather than genuine state preparation; Section 7 addresses this, for a restricted but practically important class of problems, with a different construction.

$$\langle \tau(x) |_n \otimes \langle \tau(x) |_n = \langle \tau(x) |_{n+1} \mathfrak{m}_1, \quad \mathfrak{m}_1 \in \mathbb{R}^{2N \times N^2}. \quad (3.13)$$

3.6 Critical Assessment

4 A Regime-Independent Quantum Pipeline

4.1 Overview and Model Configuration

Pipeline schematic (TikZ). Problem specification (equation, constraints, variant, n) $\rightarrow$ residual assembly (standard form | descriptor form) $\rightarrow$ stacked-residual block encoding $U_R \rightarrow$ reflection $V \rightarrow$ walk $W \rightarrow$ GQSP imaginary-time filter $\rightarrow$ post-selection $\rightarrow$ readout. Alongside: the classical reference (SVD of R) and the three verification tiers (circuit simulation, exact emulation, resource estimate) attached to the stages they check.

Figure 2: Model configuration: from problem specification to decoded field.

4.2 Problem Specification and Model Inputs

Table 2: The shared constraint layer: every constraint row, in either form, is one of four kinds.

Kind	Operator	Use
Zero (invariant)	$\mathbb{B}(x)$ on the chosen field or derivative block	$f(x_z) = 0, f'(x_m) = 0$
Collapse (regular)	$\mathbb{D}^{(0)}(x_s)$	constants, sources; lifting linear terms in the doubled space
Ratio (vector data)	$\mathbb{F} = y_a(I - a\rangle\langle a) - w\langle a , \quad w = \sum_{c \neq a} y_c c\rangle$	multi-component initial data (Carleman); coupled fields
Datum slice	$(\mathbb{S}_a(x^*) - \hat{g} \tau(x_s)^\top / v) \Pi_f$	inhomogeneous boundary or initial functions, without doubling

4.3 The Classical Reference

$$|\psi_{\text{ref}}\rangle = \arg \min_{\|\psi\|=1} \|R\psi\|_2, \quad \kappa(R^\dagger R) = \kappa(R)^2. \quad (4.1)$$

Table 3: The benchmark ladder. Each problem adds one capability to those below it; R2–R6 are the paper’s Figs 3–7.

	Problem	Reference	New capability
R1	$f' + 2f = 0, f(-1) = 1$	$e^{-2(x+1)}$	smallest end-to-end system
R2	Fig. 3: $\{1, 4, 4\}, \{1, -2, -3\}, \{1, 5, 400\}$	closed form	order 2; Dirichlet and Neumann invariants; stiffness
R3	Fig. 4: Legendre, $m = 0$ ($l = 4, 5$) and $m = 1$ ($l = 5, 6$)	P_l, P_l^1	variable coefficients and lifts; an exact kernel ($m = 0$)
R4	Fig. 5: four inhomogeneous equations	closed form	source injection via $\mathbb{D}^{(0)}$; Maclaurin sources
R5	Fig. 6: Laplace (2-D), heat and wave (1+1-D)	closed form	multi-axis tensors; invariants versus datum slices
R6	Fig. 7: $4f'' + 2f'^2 + f = 0; f'' - 2f^2 + x = 0$	$1 - x^2/8$; boundary-value solve	doubled space (kernel degeneracy)
R7	$u' = v, v' = -\omega^2 u, \omega = \pi$	sin, cos	two fields; ratio rows
R8	$y' = -y + y^2, y(0) = \frac{1}{2}$	$1/(1 + e^t)$	Carleman lift, both forms in time
R9	viscous Burgers, periodic, Fourier–Galerkin	Cole–Hopf; DNS	field-scale Carleman; convolution F_2 ; QFT fold
R10	2-D vorticity Navier–Stokes, periodic	Taylor–Green; DNS	the culmination

4.4 From Stacked Residual to Quantum Walk

$$R = \begin{pmatrix} R_1 \\ \vdots \\ R_L \end{pmatrix}, \quad (\langle j | \otimes \langle 0_a |) U_R (|0\rangle \otimes |0_a\rangle) = \frac{R_j}{\alpha_R}, \quad \alpha_R = \left(\sum_{j=1}^L \alpha_j^2 \right)^{1/2}, \quad H = R^\dagger R. \quad (4.2)$$

Proposition 1 (Stacked-residual reflection).

$$V = U_R^\dagger (2\Pi_{a=0} - I) U_R, \quad V^\dagger = V, \quad V^2 = I, \quad (4.3)$$

$$\Pi_{\text{in}} V \Pi_{\text{in}} = \frac{2H}{\alpha_R^2} - I \equiv H', \quad C(V) = U_R^\dagger C(2\Pi_{a=0} - I) U_R.$$

$$W = (2\Pi_{\text{in}} - I) V, \quad \Pi_{\text{in}} W^k \Pi_{\text{in}} = T_k(H'). \quad (4.4)$$

4.5 GQSP Imaginary-Time Filtering with A Priori Parameters

$$\Pi_{\text{in}} P(W) \Pi_{\text{in}} = \sum_{k=0}^d q_k T_k(H'), \quad P(z) = \sum_{k=0}^d q_k z^k, \quad \sup_{|z|=1} |P(z)| \leq 1. \quad (4.5)$$

$$f_\beta(x) = e^{-\beta(1+x)}, \quad f_\beta(H') = e^{-2\beta H/\alpha_R^2}, \quad q_k = (2 - \delta_{k0})(-1)^k e^{-\beta} I_k(\beta). \quad (4.6)$$

Proposition 2 (Normalisation of the imaginary-time filter).

$$\sum_{k=0}^{\infty} |q_k| = 1. \quad (4.7)$$

$$\beta = \frac{\alpha_R^2}{2\Delta} \ln \frac{1}{\varepsilon_f}, \quad d \approx \sqrt{2\beta \ln(2/\varepsilon_t)}, \quad (4.8)$$

$$d = \Theta\left(\sqrt{\frac{\alpha_H}{\Delta}} \log \frac{1}{\varepsilon}\right), \quad \alpha_H = \alpha_R^2. \quad (4.9)$$

$$|\varphi_0\rangle \propto \bigotimes_{j=0}^{n-1} \left(|0\rangle + r^{2^j} |1\rangle\right), \quad \gamma = |\langle\psi_{\text{ref}}|\varphi_0\rangle|. \quad (4.10)$$

$$p_G = \|f_\beta(H') \varphi_0\|^2 \rightarrow \gamma^2, \quad \text{gates} \approx \frac{\pi}{4\gamma} \left[2d g(U_R) + d g(\text{refl})\right]. \quad (4.11)$$

Optional quantikz circuit: one GQSP ancilla; $R_0 \prod_{k=1}^d [C_0 W R_k]$ with $C_0 W$ the controlled walk built from U_R , $U_R^\dagger$ and the two reflections, only the reflections controlled (Proposition 1); post-selection on the ancilla and the block-encoding register.

Figure 3: The GQSP imaginary-time circuit.

4.6 Reading Out the Solution

$$f^*(x) = 4P_C(x) - P_G(x) - \frac{1}{2N} = \sqrt{\frac{p_G}{N}} \tau(x) \cdot \psi, \quad (4.12)$$

$$f_Q(x) = f(x_s) \frac{f^*(x)}{f^*(x_s)}, \quad \eta_e = \frac{f(x_s)^2 p_G}{N f^*(x_s)^2}. \quad (4.13)$$

4.7 Verification Framework and Error Budget

$$\|f_Q - f\| \leq \underbrace{\|f_{\text{ref}} - f\|}_{\varepsilon_{\text{disc}}} + \kappa_{\text{read}} (\varepsilon_{\text{filter}} + \varepsilon_{\text{trunc}} + \varepsilon_{\text{phase}} + d \varepsilon_{\text{BE}}) + \varepsilon_{\text{sample}}. \quad (4.14)$$

Table 4: Operator-verification modes: what each compares, up to what size, and to what tolerance.

Mode	Comparison	Size	Tolerance
Exact	statevector per input column; $\alpha \Pi_{\text{out}} U \Pi_{\text{in}}$ against the independently assembled operator	$\lesssim 20$ qubits	max-abs $\leq 10^{-10} \ X\ $
Probe	16–32 Haar-random and structured in- puts against $X\psi/\alpha$ by sparse matrix- vector products	$\lesssim 28\text{--}30$ qubits	relative $\leq 10^{-9}$
IR	compiled term IR against the residual assembly at every size; each term type exactly at small n	any n	$\leq 10^{-12}$
Approximation	published $\mathbb{G}$ only: entry error $\epsilon_{\mathbb{G}}(n)$ against the small-angle bound	exact sizes	reported, not gated

Table 5: Acceptance criteria for every state-preparation run.

Level	Metric	Acceptance
State	infidelity $1 - \langle \psi_{\text{ref}} \psi \rangle ^2$ in the form’s own space	$\leq \varepsilon = 10^{-6}$, and $\leq 10\times$ the predicted bound
Probability	measured p_G against $\ P(H')\varphi_0\ ^2$	relative difference $\leq 10^{-8}$
Field	decoded f_Q against the analytic solution (L^∞ , relative L^2)	$\leq \varepsilon_{\text{disc}} + \kappa_{\text{read}} \sqrt{\varepsilon_{\text{prep}}}$
T1 against T2	state difference	$\leq 10^{-10}$
Phases	realised against target polynomial	$\leq 10^{-12}$

4.8 Software and Computational Configuration

5 The Standard Form, Encoded Exactly

5.1 The Published Encoding, Rebuilt

5.2 An Exact Factorisation of the Differentiation Matrix

Proposition 3 (Exact factorisation of the differentiation matrix).

$$\begin{aligned} \mathbb{G} &= R_0 (2U_{\text{odd}}) \Lambda, \\ R_0 &= I - \left(1 - \frac{1}{\sqrt{2}}\right) |0\rangle \langle 0|, \quad (U_{\text{odd}})_{kp} = [p > k, p - k \text{ odd}], \quad \Lambda = \text{diag}(p). \end{aligned} \quad (5.1)$$

5.3 Circuits and Cost

$$\Lambda = \frac{N-1}{2} I - \sum_{j=0}^{n-1} 2^{j-1} Z_j. \quad (5.2)$$

Quantikz circuit: offset register (PREPARE: X on the lowest bit, H on the rest), system register p , borrow-flag ancilla; a ripple-carry subtractor $p \leftarrow p - d$ [3] whose borrow flag is post-selected on $|0\rangle$, so every wrapped term vanishes exactly.

Figure 4: SELECT for the odd-offset shifts: in-place subtraction with a borrow flag.

Proposition 4 (Cost of the structured encoding).

$$\alpha_{\mathbb{G}} = N(N-1), \quad \frac{\alpha_{\mathbb{G}}}{\|\mathbb{G}\|_2} \leq 2.12 \quad (n \leq 10), \quad O(n^2) \text{ gates}, \quad n + O(\log n) \text{ ancilla.} \quad (5.3)$$

Table 6: Subnormalisation of the differentiation-matrix encodings, relative to $\|\mathbb{G}\|_2$.

n	$\ \mathbb{G}\ _2$	Structured (Proposition 4)	Published (Equation (3.11))
2	7.39	1.623	49.9
4	121	1.986	4.05×10^3
7	7739	2.100	2.20×10^6
10	495 308	2.115	1.14×10^9

5.4 What Remains: The Standard-Form Hamiltonian

$$\|H_{\text{std}}\| = \Theta(N^8), \quad \Delta = \Theta(1) \implies d = \tilde{\Theta}(N^4). \quad (5.4)$$

6 The Descriptor Reformulation

6.1 The Idea

The differentiation operator $\mathbb{G}$ is dense. Generally, dense operators are expensive to block-encode and sparse matrices are cheap.

Two sparsity (spy) plots side by side for $\{1, 4, 4\}$ (R2a) at $n = 4$: left, $A_{\text{std}} = \mathbb{G}^2 + 4\mathbb{G} + 4I$ with its dense upper triangle; right, A_{desc} with its block-banded pattern (recurrence blocks, equation row, padding). Generated from `pihm.assemble`.

Figure 5: Dense against banded: the two residuals of the same equation.

6.2 The Banded Factorisation

Differentiation of a Chebyshev series is a sparse operation written densely. Let $f = \sum_{\ell} \hat{f}_{\ell}^{(0)} T_{\ell}$ have derivative $f' = \sum_{\ell} \hat{f}_{\ell}^{(1)} T_{\ell}$; for $\ell \geq 2$ the coefficient vectors $\hat{f}^{(0)}$ and $\hat{f}^{(1)}$ are tied by the recurrence

$$\hat{f}_{\ell-1}^{(1)} - \hat{f}_{\ell+1}^{(1)} = 2\ell \hat{f}_{\ell}^{(0)}, \quad (6.1)$$

with the $\ell = 1$ row carrying an extra factor of 2 on $\hat{f}_0^{(1)}$, a normalisation artefact of T_0 that Appendix B states in full and derives. Each row relates the coefficient $\hat{f}_{\ell}^{(0)}$ to just two coefficients of the derivative. In the normalised Chebyshev basis this is a matrix identity

$$\tilde{\mathbb{B}} \hat{f}^{(1)} = \tilde{\mathbb{D}} \hat{f}^{(0)}, \quad (6.2)$$

with $\tilde{\mathbb{B}}$ banded (its main diagonal and its second superdiagonal), and $\tilde{\mathbb{D}}$ carrying a single superdiagonal, such that both hold $\mathcal{O}(N)$ non-zero entries. The dense Chebyshev derivative matrix $\mathbb{G}$ is the operator that sends a function's coefficients directly to those of its derivative, $\hat{f}^{(1)} = \mathbb{G} \hat{f}^{(0)}$. A simple rearranging of Equation (6.2) recovers it as the inverse

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1} \tilde{\mathbb{D}}, \quad (6.3)$$

Add a reference for this

Remember to define dense vs sparse as part of background

an identity that holds exactly. The density of $\mathbb{G}$ is due to inverting a banded matrix, and Equation (6.2) is the same linear map before that inverse is done. This is the discrete form of the ultraspherical spectral method [4], in which differentiation becomes sparse once the derivative is allowed to sit in a polynomial basis one order above the function.

6.3 The Descriptor Residual and Kernel Equivalence

Consider the homogeneous order- k linear ordinary differential equation (ODE)

$$\sum_{j=0}^k a_j(x) f^{(j)}(x) = 0, \quad x \in [-1, 1], \quad (6.4)$$

where $a_j(x)$ are the (possibly x -dependent) coefficients of each j^{th} derivative $f^{(j)}$.

We discretise by representing every function appearing in the equation by its first N coefficients in the normalised Chebyshev basis. Concretely, let $\hat{f}^{(j)} \in \mathbb{C}^N$ denote the coefficient vector of $f^{(j)}$, for $j = 0, \dots, k$ and stack these into a single vector,

$$w = (\hat{f}^{(0)}; \hat{f}^{(1)}; \dots; \hat{f}^{(k)}). \quad (6.5)$$

Two kinds of linear relations couple the blocks $\hat{f}^{(j)}$:

Recurrence rows. The coefficients of consecutive derivatives are related by

$$-\tilde{\mathbb{D}} \hat{f}^{(j)} + \tilde{\mathbb{B}} \hat{f}^{(j+1)} = 0, \quad j = 0, \dots, k-1. \quad (6.6)$$

This gives k rows of the system, each enforcing the derivative relationship between $\hat{f}^{(j+1)}$ and $\hat{f}^{(j)}$. These rows are algebraic since they carry no derivative and only fix each $\hat{f}^{(j+1)}$ as the derivative of $\hat{f}^{(j)}$, being Equation (6.2) applied block-wise and hence equivalent to $\hat{f}^{(j+1)} = \mathbb{G} \hat{f}^{(j)}$ on $\ker A_{\text{desc}}$.

Differential row. The differential equation itself becomes one further row, obtained by multiplying each coefficient vector $\hat{f}^{(j)}$ by the (possibly x -dependent) coefficient $a_j(x)$ and summing:

$$\sum_{j=0}^k \mathbb{M}_{a_j} \hat{f}^{(j)} = 0, \quad (6.7)$$

where $\mathbb{M}_{a_j}$ is the matrix representing multiplication by $a_j(x)$ in coefficient space (reducing to $a_j I_N$ when a_j is constant). This single row is the equation itself: we carry the intermediate derivatives as their own unknown coefficient vectors within w , rather than representing them via $\hat{f}^{(j)} = \mathbb{G}^j \hat{f}^{(0)}$, so that Equation (6.3) (the only dense step) never appears.

Together these $k+1$ block rows (k recurrence rows and one differential row) define a linear system for w and transform the original ODE problem into a differential-algebraic equation

(DAE) system.

Definition 11 (Descriptor form residual operator): Collecting all block rows, the descriptor form chebyshev residual operator A_{desc} is the $(k+1)N \times (k+1)N$ block matrix

$$A_{\text{desc}} = \begin{pmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & & \\ & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ \mathbb{M}_{a_0} & \mathbb{M}_{a_1} & \cdots & \cdots & \mathbb{M}_{a_k} \end{pmatrix}, \quad (6.8)$$

where the first k rows are the recurrence rows and the $(k+1)$ -th row is the differential row, so that the whole system is compactly written as

$$A_{\text{desc}} w = 0. \quad (6.9)$$

A first-order example. For $f' + 2f = 0$ the state is $w = (\hat{f}^{(0)}; \hat{f}^{(1)})$, and the two facts relating its blocks are the recurrence $\tilde{\mathbb{B}}\hat{f}^{(1)} - \tilde{\mathbb{D}}\hat{f}^{(0)} = 0$ and the equation $2\hat{f}^{(0)} + \hat{f}^{(1)} = 0$, giving

$$A_{\text{desc}} = \begin{pmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ 2I_N & I_N \end{pmatrix}, \quad w = \begin{pmatrix} \hat{f}^{(0)} \\ \hat{f}^{(1)} \end{pmatrix}. \quad (6.10)$$

Every block is $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$ or a multiple of the identity, so A_{desc} is banded, against the dense $\mathbb{G} + 2I$ that substituting $\hat{f}^{(1)} = \mathbb{G}\hat{f}^{(0)}$ would produce. The fully populated $N = 4$ instance of Equation (6.10), the entries of $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$, and the exactness of Equation (6.3), are given in Appendix B.

Table 7: The block rows of the descriptor residual A_{desc} for an order- k equation.

Block row	Equation
$j = 0, \dots, k-1$ (recurrence)	$-\tilde{\mathbb{D}} \hat{f}^{(j)} + \tilde{\mathbb{B}} \hat{f}^{(j+1)} = 0$
k (the equation)	$\sum_j \mathbb{M}_{a_j} \hat{f}^{(j)} - \mathbb{M}_r \mathbb{D}^{(0)} \hat{f}^{(0)} = 0$
$b > k$ (padding)	$c_{\text{pad}} \hat{f}^{(b)} = 0$

Theorem 1 (Kernel equivalence).

$$\ker A_{\text{desc}} = \{ (p, \mathbb{G}p, \dots, \mathbb{G}^k p) : p \in \ker A_{\text{std}} \}. \quad (6.11)$$

Cost. The reformulation trades a higher dimensional Hilbert space (and with it a higher qubit count) for a sparser residual operator. w has $(k+1)N$ components at the classical, unpadding level; realising the field register on $\lceil \log_2(k+1) \rceil$ qubits pads this to $2^{\lceil \log_2(k+1) \rceil} N$ (Appendix G.2), which is what should be compared against the N of the standard form. Each block of A_{desc} is $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$, or a banded multiplication operator (bandwidth q for a degree- q polynomial coefficient;

Appendix G.2), so A_{desc} carries $\mathcal{O}(1)$ occupied diagonals per block, whereas the standard-form residual $A_{\text{std}} = \sum_j a_j \mathbb{G}^j$ is dense with $\Theta(N^2)$ non-zero entries. The $\hat{f}^{(0)}$ -block of the recovered null vector is the Chebyshev coefficient vector the standard form solves for, agreeing with it to within numerical precision, quantified in Section 8. Carrying the intermediate derivatives instead of eliminating them converts the dense physics-informed residual into a banded one, and every later stage of the pipeline inherits that structure.

6.4 Constraints, Partial Differential Equations and Coupled Fields

6.5 Block Rescaling

$$\hat{f}^{(m)} \mapsto \zeta^{-m} \hat{f}^{(m)}, \quad m = 0, \dots, k. \quad (6.12)$$

6.6 The Block Encoding: One Flat Linear Combination

$$\text{diag}(k)_{k=0}^{N-1} = \sum_{j=0}^{n-1} 2^j \frac{I - Z_j}{2}, \quad R_m = I - 2|m\rangle\langle m|. \quad (6.13)$$

$$A_{\text{desc}} = \sum_{t=1}^L c_t U_t, \quad U_t = (\text{field word}) \otimes \text{SHIFT}^{s_t} P_t, \quad \alpha_A = \sum_{t=1}^L |c_t|. \quad (6.14)$$

Quantikz circuit: selector register (PREPARE loads $\sqrt{|c_t|/\alpha_A}$), field register (n_f qubits), system register (n qubits); SELECT applies each term's field word, cyclic shift SHIFT^{s_t} (multi-controlled-X ladder) and Z-string or reflection P_t ; $\text{PREPARE}^\dagger$; post-selection on the selector.

Figure 6: The descriptor block encoding: one PREPARE/SELECT/PREPARE[†] over the whole residual.

Input: the term IR $\{(c_t, U_t)\}_{t=1}^L$ of A_{desc} (Equation (6.14))

Output: PREPARE and SELECT circuits; the subnormalisation α_A

$\alpha_A \leftarrow \sum_t |c_t|$;

PREPARE: $|0\rangle \mapsto \sum_t \sqrt{|c_t|/\alpha_A} |t\rangle$;

foreach term t **do**

 | SELECT: controlled on $|t\rangle$, apply $\text{sgn}(c_t) U_t$;

end

return PREPARE, SELECT, α_A ;

Algorithm 1: Compiling the term IR of A_{desc} into PREPARE and SELECT.

Table 8: The atoms of the descriptor block encoding.

Atom	Arity	Gates	Ancilla	Verified residual
Cyclic shift SHIFT ^s				
Z-string diagonal				
Reflection R_m				
Feature-map rank-one row				
Block-rescaling diagonal				

6.7 Cost of the Descriptor Encoding

Proposition 5 (Cost of the descriptor encoding).

$$\alpha_A = 2N + O(1), \quad \frac{\alpha_A}{\|A_{\text{desc}}\|_2} \rightarrow 1, \quad \lceil \log_2(n+4) \rceil + 6 \text{ ancilla}, \quad O(n^2) \text{ gates.} \quad (6.15)$$

6.8 Consequences for the Spectrum and the Filter

$$\|H_{\text{desc}}\| = \Theta(N^2), \quad \frac{\alpha_H}{\|H_{\text{desc}}\|} \rightarrow 1 \implies d = \tilde{\Theta}(N). \quad (6.16)$$

6.9 Summary

7 Polynomial Nonlinearity

7.1 Two Regime-Independent Treatments

7.2 The Doubled Space: Exact Encoding, Degenerate Kernel

The doubled Hilbert space method can create degenerate kernels. Quantum imaginary time evolution (QITE) can't select the physical solution from amongst a set of degenerate ground states, so although it is possible to encode the solution to the original problem in the ground state of the Hamiltonian, it is not possible to extract it from the ground state of the descriptor Hamiltonian.

Proposition 6 (Imaginary time projects onto a degenerate kernel).

$$\lim_{\beta \rightarrow \infty} \frac{e^{-\beta H_{\otimes}} \varphi_0}{\|e^{-\beta H_{\otimes}} \varphi_0\|} = \frac{\Pi_{\ker H_{\otimes}} \varphi_0}{\|\Pi_{\ker H_{\otimes}} \varphi_0\|}, \quad \Psi = \psi \otimes \psi. \quad (7.1)$$

7.3 The Carleman Lift in Space–Time

$$\frac{du}{dt} = \sum_{p=1}^P F_p u^{\otimes p} \longrightarrow \frac{dz}{dt} = \mathbb{A}_K z, \quad z = (u, u^{\otimes 2}, \dots, u^{\otimes K}), \quad (7.2)$$

$$\rho = \frac{\|u_0\| \|F_2\|}{|\operatorname{Re} \lambda_1(F_1)|} < 1, \quad \text{truncation error} = O(\rho^K), \quad (7.3)$$

$$\begin{aligned} A_{\text{st}}^{\text{std}} &= \mathbb{G}_t \otimes I - \frac{T}{2} I \otimes \mathbb{A}_K, \\ A_{\text{st}}^{\text{desc}} &= \tilde{\mathbb{D}}_t \otimes I - \frac{T}{2} \tilde{\mathbb{B}}_t \otimes \mathbb{A}_K = (\tilde{\mathbb{B}}_t \otimes I) A_{\text{st}}^{\text{std}}, \end{aligned} \quad \xi = \frac{2t}{T} - 1. \quad (7.4)$$

7.4 Product Folds on a Circuit

Proposition 7 (The nodal fold is optimal).

$$\mathfrak{m}_1^{(p)} = 2^{p\ell/2} \mathbb{U}^\dagger \Delta_p \mathbb{U}^{\otimes p}, \quad \alpha(\mathfrak{m}_1^{(2)}) = 2 = \|\mathfrak{m}_1^{(2)}\|. \quad (7.5)$$

7.5 Fourier–Galerkin Fluids: Burgers and Navier–Stokes

$$u_t + u u_x = \nu u_{xx}, \quad F_1 = \text{diag}(-\nu k^2), \quad (7.6)$$

$$\omega_t + J(\psi, \omega) = \nu \nabla^2 \omega, \quad \nabla^2 \psi = -\omega \text{ on } [0, 2\pi)^2, \quad \omega_0^{\text{TG}} = 2 \sin x \sin y. \quad (7.7)$$

7.6 Preparation Versus Projection

Table 9: Which nonlinear route prepares the solution, and which only projects onto a kernel that contains it.

Route	Kernel	What imaginary time returns	Precondition
Doubled space (Wu et al.; R6)	degenerate, dimension $\geq 2^{2n-1}$	the projection of φ_0 onto the kernel	none
Carleman lift (R8–R10), either form	one-dimensional near-kernel	the truncated solution	$\rho < 1$

8 Numerical Study

8.1 Experimental Design and Cost Conventions

8.2 Reproducing Wu et al.

$$\eta_e = \|b\|^2 = \frac{N}{\pi} \int_{-1}^1 \frac{f(x)^2}{\sqrt{1-x^2}} dx. \quad (8.1)$$

Table 10: The paper’s printed scale factors against the exact solution’s $\|b\|^2$ (Equation (8.1)) and the paper-faithful ground state at the printed n . † : the ground state is not trusted (a degenerate kernel, or a gap ratio below 100). Classical computation.

Fig.	Problem	n	Printed η_e	Ideal $\ b\ ^2$	Ground state	Agreement
3b	R2a	3	1.29	1.29009	1.28731 †	✓
3c	R2b	3	32.47	32.47741	32.22636 †	✓ (one unit low)
3d	R2c	5	451.71	451.70962	451.52312	✓
4a	R3a	3	1.75	1.74512	1.74512	✓
4b	R3b	3	1.49	1.48779	1.48779	✓
4c	R3c	5	108.51	108.51142	109.70433 †	✓
4d	R3d	5	135.38	135.38360	137.46159 †	✓
5a	R4a	3	1.35	1.34649	1.34478	✓
5b	R4b	3	14.74	14.74797	14.72531 †	✓ (one unit low)
5c	R4c	4	0.82	0.82116	0.82521	✓
5d	R4d	4	77.51	77.51167	76.93737	✓
6b	R5a	3	5.21559	5.21653	5.24407 †	×
6d	R5b	4	603.863	598.86606	598.87966	×
6f	R5c	5	216.469	239.82712	239.82712	×
7b	R6a	2	3.75367	3.52344	3.52344 †	×
7c	R6b	2	0.07984	0.04383	0.04182 †	×

Four panels, the paper’s Figs 5a–5d (R4a–R4d): the analytic solution, the paper-faithful ground state (rounded constraint points, Maclaurin sources) and the corrected ground state (exact roots, Chebyshev sources) at the printed n ; inset or second row: L^∞ error against n for both variants. Source: notebook *10_paper_reproduction*. Classical computation.

Figure 7: Printed choices distort the paper’s solutions at the printed resolution.

Verification error against n for the published, structured and descriptor encodings: exact mode (filled markers), probe mode (open markers), IR mode (crosses), with each mode’s tolerance as a dashed line; a second panel gives transpiled CNOT count per query against n with fitted exponents. Source: notebook *20_pillar1_operators*.

Figure 8: Every encoding is verified to its mode’s tolerance at every size tested.

Table 11: Where the cost lies: the paper’s regime under its published encoding and under the structured exact encoding, against the descriptor form, for Fig. 3b ($\{1, 4, 4\}$, $f(-1) = 0$). Tier: T2 where measured, T3 beyond.

Quantity	Standard, published	Standard, structured	Descriptor
$\ H\ $	$\Theta(N^8)$	$\Theta(N^8)$	$\Theta(N^2)$
Gap Δ	7.04 (flat)	7.04 (flat)	0.454 (flat)
$\alpha_{\mathbb{G}}/\ \mathbb{G}\ _2$	$\sim N^3/2$	≤ 2.12	no $\mathbb{G}$
$\alpha_H/\ H\ $	$\sim N^{12}$	$22 \rightarrow 60$	$5.3 \rightarrow 1.05$
Gates per query	$O(n^3)$, approximate	$O(n^2)$, exact	$O(n^2)$, exact
Ancilla	$2n + 3$ ($\mathbb{G}$ alone)	$n + O(\log n)$ per $\mathbb{G}$	$\lceil \log_2(n + 4) \rceil + 6$
Degree at $n = 8$ ($\varepsilon = 10^{-6}$)	8.2×10^{23}	1.2×10^{10}	5.6×10^3
Degree scaling	$\sim N^{10}$	$\tilde{\Theta}(N^4)$	$\tilde{\Theta}(N)$

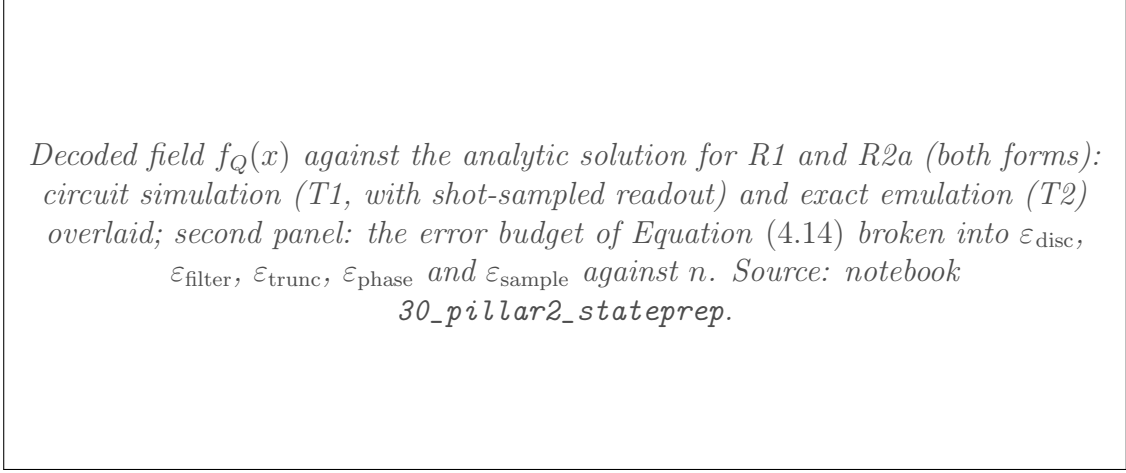


Figure 9: The prepared state reproduces the analytic solution to within the predicted error budget.

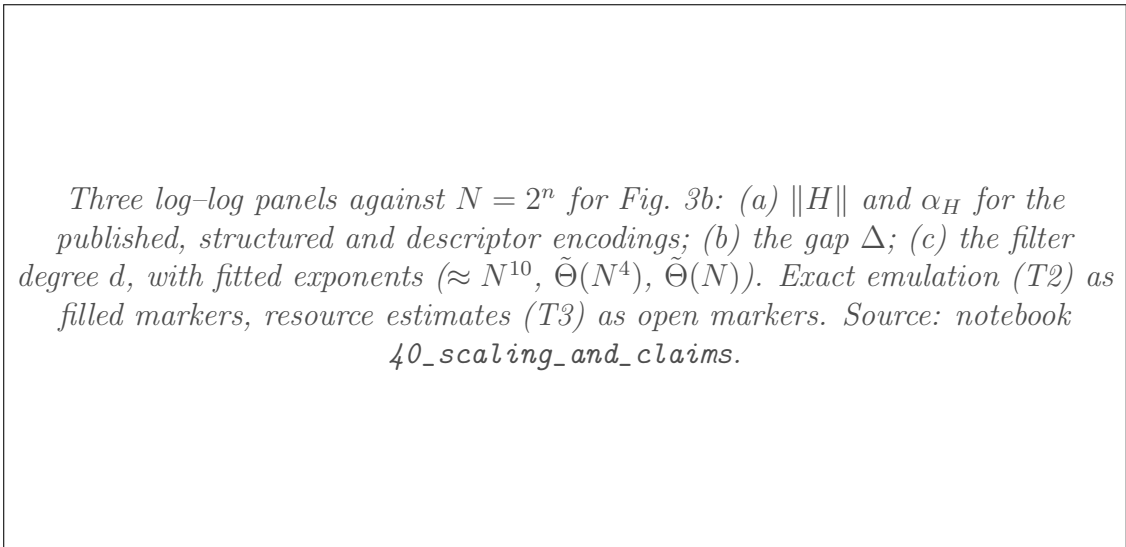


Figure 10: The encoder wall and the Hamiltonian wall are separate gaps.

Three columns (R2c, R3a, R5b or R5c): top row, the decoded solution against the analytic one; bottom row, error against n for the standard and descriptor forms (and, for R2c, with and without block rescaling). Exact emulation (T2). Source: notebook 40_scaling_and_claims.

Figure 11: Three representative problems, solved in both forms.

(a) R8: error against Carleman order K for several time resolutions n_t , both forms; (b) R6: projection fidelity against physical fidelity; (c) R10: Taylor–Green and vortex-pair vorticity snapshots against DNS, with the tier marked (T2 small, T3 beyond). Source: notebook 50_navier_stokes.

Figure 12: Carleman truncation converges geometrically once time is resolved; the doubled space only projects.

8.3 Operator Verification and Encoding Cost

8.4 State Preparation and Readout

8.5 Scaling: Where the Cost Lies

8.6 Representative Problems

8.7 Nonlinear Benchmarks

8.8 Case Study: Two-Dimensional Navier–Stokes

8.9 Summary of Findings

9 Discussion

9.1 Why the Descriptor Form Wins, and What It Costs

9.2 End-to-End Complexity

9.3 What the Reproduction Says About the Published Method

9.4 Where the Construction Applies

9.5 Feasibility on Fault-Tolerant Hardware

9.6 Limitations

10 Conclusions and Future Work

10.1 Conclusions

10.2 Future Work

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A Quantum Gate Reference

Table 12: The gates used in this thesis.

Gate	Circuit symbol	Matrix
Hadamard		
Pauli X, Y, Z		
Rotation $R_Y(\theta)$		
Phase $P(\phi)$		
CNOT		
Toffoli		
Multi-controlled X		
Zero-state reflection $2 0\rangle\langle 0 - I$		
Quantum Fourier transform		

B Chebyshev and Ultraspherical Identities

This appendix records the entries of the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ introduced in Section 6.2, the exact factorisation $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$, and the fully populated $N = 4$ instance of the first-order descriptor operator.

B.1 The differentiation recurrence

For $f = \sum_{\ell \geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell \geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1} p_{\ell-1} - p_{\ell+1} = 2\ell a_\ell, \quad \ell \geq 1, \quad (\text{B.1})$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on p_0 . Truncating to $N = 2^n$ coefficients and dropping the p_N term, Equation (B.1) is the matrix identity $\tilde{\mathbb{B}}_{\text{raw}} p = \tilde{\mathbb{D}}_{\text{raw}} a$ with $\tilde{\mathbb{D}}_{\text{raw}}$ a single superdiagonal, $(\tilde{\mathbb{D}}_{\text{raw}})_{\ell, \ell+1} = 2(\ell + 1)$, and $\tilde{\mathbb{B}}_{\text{raw}}$ two diagonals, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell, \ell} = c_\ell$, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell, \ell+2} = -1$.

B.2 The normalised basis

The construction of Section 6.2 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ ($\ell \geq 1$), orthonormal under the discrete N -point Chebyshev inner product $\langle T_k, T_\ell \rangle_N = \frac{N}{2} c_\ell \delta_{k\ell}$ [2]: since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \text{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{\mathbb{B}} = S \tilde{\mathbb{B}}_{\text{raw}} S^{-1}$ and $\tilde{\mathbb{D}} = S \tilde{\mathbb{D}}_{\text{raw}} S^{-1}$, that is

$$\tilde{\mathbb{D}}_{\ell, \ell+1} = 2(\ell + 1)\sqrt{c_\ell/c_{\ell+1}}, \quad \tilde{\mathbb{B}}_{\ell, \ell} = c_\ell, \quad \tilde{\mathbb{B}}_{\ell, \ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \quad (\text{B.2})$$

so $\tilde{\mathbb{D}}$ has the single non-zero diagonal $(2\sqrt{2}, 4, 6, 8, \dots)$ and $\tilde{\mathbb{B}}$ has main diagonal $(2, 1, 1, \dots)$ and second superdiagonal $(-\sqrt{2}, -1, -1, \dots)$. Both hold $\mathcal{O}(N)$ non-zero entries at any N .

The dense Chebyshev derivative matrix is their exact quotient,

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1} \tilde{\mathbb{D}}, \quad (\text{B.3})$$

an algebraic identity rather than a numerical approximation: $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ are read directly off the same recurrence that defines $\mathbb{G}$ (Equation (B.1)), so the equality holds exactly at every N , independent of whichever values a test suite happens to check. Appendix I reports this identity's residual against the reference implementation, alongside the rest of the composition-algebra checks.

B.3 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [4]. Stating the correspondence precisely is what fixes what this thesis adds to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis T_k but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1 - x^2)^{\lambda-1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{dT_k}{dx} = k C_{k-1}^{(1)}, \quad k \geq 1 \quad (\text{B.4})$$

[4, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of u' in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because u' 's coefficients now live in a different basis to u 's own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ ($k = 0$), $\frac{1}{2}$ ($k \geq 1$), and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [4, Eq. 2.8]. Comparing entries, $\tilde{\mathbb{D}}_{\text{raw}} = 2\mathcal{D}_0$ and $\tilde{\mathbb{B}}_{\text{raw}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{\mathbb{B}}_{\text{raw}} p = \tilde{\mathbb{D}}_{\text{raw}} a$ of Equation (B.1) is literally $\mathcal{S}_0 p = \mathcal{D}_0 a$, doubled by the convention that folds the weight of T_0 into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow the framing of Section 6.2 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that p 's own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for p via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \geq 1$ in [4] raises the basis one level per derivative: a λ -th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $dC_k^{(\lambda)}/dx = 2\lambda C_{k-1}^{(\lambda+1)}$ ($k \geq 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \rightarrow C^{(\lambda+1)}$, still two diagonals [4, Eqs. 3.2–3.3]. To assemble one equation of order k , every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1} \cdots \mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda] \mathcal{D}_\lambda$, giving one almost-banded operator $L = \mathcal{M}_k[a^k] \mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1} \cdots \mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda] \mathcal{D}_\lambda$ that is solved once for the coefficients of $u^{(k)}$ [4, §3.2] – so a k -th order equation needs k distinct conversion

operators $\mathcal{S}_0, \dots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of Section 6.3 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $\hat{f}^{(j)}$ of w , and reuses the single $\lambda = 0$ pair $(\tilde{\mathbb{B}}, \tilde{\mathbb{D}})$ between every consecutive pair of blocks (Equation (6.6)). No elimination is performed, and no operator beyond $\tilde{\mathbb{B}}, \tilde{\mathbb{D}}$ and the multiplication operators of Appendix G.2 is required at any order k – the price is the larger state w that carries every intermediate derivative explicitly instead of eliminating it (Appendix G.2). What this thesis adds to [4] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [1], whose block structure is exactly what Section 6.6 block-encodes.

B.4 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (B.2) gives

$$\tilde{\mathbb{B}} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tilde{\mathbb{D}} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbb{G} = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B.5})$$

The top-right entry $3\sqrt{2}$ of $\mathbb{G}$ is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row ℓ of $\mathbb{G}$ is non-zero at columns $\ell + 1, \ell + 3, \dots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{\mathbb{B}}$ or $\tilde{\mathbb{D}}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ of Section 6.3, $w = (\hat{f}^{(0)}; \hat{f}^{(1)})$ and $A_{\text{desc}} = \begin{pmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ 2I_4 & I_4 \end{pmatrix}$ is the 8×8 matrix

$$A_{\text{desc}} = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array} \right]. \quad (\text{B.6})$$

The top four rows are the recurrence block $-\tilde{\mathbb{D}} \hat{f}^{(0)} + \tilde{\mathbb{B}} \hat{f}^{(1)} = 0$; the bottom four are the differential row $2\hat{f}^{(0)} + \hat{f}^{(1)} = 0$. Every non-zero entry lies on a diagonal or a near-diagonal, in place of the dense $\mathbb{G} + 2I$ of the eliminated form.

C The Standard Form of Wu et al. in Full

This appendix gives the $N = 4$ instance of the standard-form residual A_{std} that Section 3.3 builds for Wu et al.’s constant-coefficient case, and the entries of the lifting operator $\mathbb{M}_{x^p}$ that the same section uses for a variable coefficient.

The standard-form residual at $N = 4$. For Wu et al.’s stiff pilot equation [1], $f'' + 5f' + 400f = 0$, Equation (3.4) gives $A_{\text{std}} = \mathbb{G}^2 + 5\mathbb{G} + 400I$. At $N = 4$, using $\mathbb{G}$ (Appendix B),

$$A_{\text{std}} = \begin{pmatrix} 400 & 5\sqrt{2} & 4\sqrt{2} & 15\sqrt{2} \\ 0 & 400 & 20 & 24 \\ 0 & 0 & 400 & 30 \\ 0 & 0 & 0 & 400 \end{pmatrix}, \quad (\text{C.1})$$

verified directly against `pihm.chebyshev_derivative_matrix`. A_{std} is triangular here only because this equation has no odd-order term beyond f' at even N ; the off-diagonal entries sit three orders of magnitude below the diagonal, a large-coefficient conditioning problem stacked on top of $\mathbb{G}$ ’s own density.

The lifting operator $\mathbb{M}_{x^p}$. $\mathbb{M}_{x^p}$ is exact and *rectangular*: multiplying a degree- $(N-1)$ model by x^p can raise its degree by as much as p , so the result is represented one register wider, $2N \times N$, with no truncation. At $n = 1$ ($N = 2$, so $p \in \{1, 2\}$),

$$\mathbb{M}_{x^1} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}, \quad \mathbb{M}_{x^2} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{3}{2\sqrt{2}} \\ \frac{1}{2} & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}, \quad (\text{C.2})$$

verified directly against `pihm.multiplication_matrix`. The general entry follows the same product-to-sum identity conjugated into the normalised bases of $|\tau(x)\rangle_n$ and $|\tau(x)\rangle_{n+1}$ that Section 3.3 states in closed form.

C.1 The paper’s printed matrices at $n = 2$

C.2 Lifts and folds

D The Published Circuits as Built

Section 3.4 describes, at the level a reader needs to follow the argument, the circuit Wu et al. use to load $\mathbb{G}$ ’s column-linear entries as a rotation amplitude [1]. This appendix gives the mechanism in full, together with a rigorous bound on its entrywise error that this thesis derives and verifies – the source paper states only that the error is “negligible” and decreases “exponentially” with n , without a closed form.

D.1 The circuits as built

D.2 The small-angle ladder and its error

The mechanism. A ladder of n singly-controlled R_y rotations acts on one flag qubit, initialised to $|1\rangle$: bit m of the (register-arithmetic-shifted) column index controls $R_y(\theta_m)$,

$$\theta_m = 2 \arcsin(-2^m \mu), \quad m = 0, \dots, n-1, \quad (\text{D.1})$$

for a small real coefficient μ . Because rotations on the same qubit compose by exact angle addition ($R_y(a)R_y(b) = R_y(a+b)$, an $SO(2)$ identity), the net rotation for a column index p with bit set S is $R_y(\Theta(p))$, $\Theta(p) = \sum_{m \in S} \theta_m$, giving the flag qubit's $|0\rangle$ -amplitude

$$\text{raw}(p) = -\sin(\Theta(p)/2) = \sin\left(\sum_{m \in S} \arcsin(2^m \mu)\right). \quad (\text{D.2})$$

The row-0 correction (entries $\sqrt{2}p$ in place of $2p$) is a second, independently accumulated ladder on the same flag qubit, with per-bit coefficient $\mu(1/\sqrt{2}-1)$, controlled on the same “this is row 0” flag the main construction already computes.

Why this is only approximate. $\text{raw}(p)$ in Equation (D.2) would equal the desired linear value μp exactly only if $\sin\left(\sum_m \arcsin(x_m)\right) = \sum_m x_m$, which does not hold in general – $\sin$ of a sum is not the sum of sines. The discrepancy is a genuine, quantifiable small-angle error.

Proposition 8 (Entrywise error of the ladder construction). *Let $J_3(p) := \sum_{m \in S} (2^m)^3$ for column index p with bit set S . Then*

$$|\text{raw}(p) - \mu p| \leq \frac{\mu^3}{6} p^3 + O(\mu^5 N^5). \quad (\text{D.3})$$

Proof. Expand both non-linearities to leading order in the small parameter μ : $\arcsin(y) = y + y^3/6 + O(y^5)$ and $\sin(x) = x - x^3/6 + O(x^5)$. Summing the first expansion over the fired bits and substituting into the second,

$$\sum_{m \in S} \arcsin(2^m \mu) = \mu p + \frac{\mu^3}{6} J_3(p) + O(\mu^5 N^5), \quad (\text{D.4})$$

$$\begin{aligned} \sin\left(\sum_{m \in S} \arcsin(2^m \mu)\right) &= \sin(\mu p) + \cos(\mu p) \cdot \frac{\mu^3}{6} J_3(p) + O(\mu^5 N^5) \\ &= \mu p + \frac{\mu^3}{6} (J_3(p) - p^3) + O(\mu^5 N^5). \end{aligned} \quad (\text{D.5})$$

Since $p = \sum_{m \in S} 2^m$ and every $2^m \leq p$, the multinomial expansion of p^3 term-by-term dominates $J_3(p)$, with equality only when S is a single bit (p itself a power of two): $J_3(p) \leq p^3$. Taking the worst case $p = N-1$ gives Equation (D.3). $\square$

This is a leading-order bound, valid once the largest single argument, $2^{n-1}\mu$, is small and well inside $\arcsin$'s domain – true from $n=2$ onward at the default $\mu = 2/\|\mathbb{G}\|_S$ (Equation (3.11)),

but not at $n = 1$, where $\mu \cdot 2^0 = \mu = 1$ exactly and the bound is not yet meaningful.

D.3 Measured resources and error

Table 13: The published encoding of $\mathbb{G}$ against the paper’s claims. Classical exact-mode measurement.

n	N	$\ \mathbb{G}\ _S$	Qubits (claim $2n + 3$)	Gates	Bound on $\epsilon_{\mathbb{G}}$	Measured $\epsilon_{\mathbb{G}}$
2	4	6.15×10^1	—	—	—	—
3	8	1.18×10^3	—	—	—	—
4	16	2.04×10^4	—	—	—	—
5	32	3.38×10^5	—	—	—	—
6	64	5.50×10^6	—	—	—	—
7	128	8.88×10^7	—	—	—	—
8	256	1.43×10^9	—	—	—	—

E The Structured Exact Encoding of the Differentiation Matrix

E.1 Proof of Proposition 3

E.2 The arithmetic SELECT

E.3 Proof of Proposition 4

E.4 Subnormalisation across n

Table 14: Subnormalisation of the structured and published encodings of $\mathbb{G}$, relative to $\|\mathbb{G}\|_2$.

n	$\ \mathbb{G}\ _2$	Structured	Published
2	7.39	1.623	49.9
3	30.1	1.858	468
4	121	1.986	4.05×10^3
5	484	2.051	3.36×10^4
6	1935	2.084	2.73×10^5
7	7739	2.100	2.20×10^6
8	30 957	2.109	1.77×10^7
9	123 827	2.113	1.42×10^8
10	495 308	2.115	1.14×10^9

F The Shared Pipeline: Identities and Formulas

F.1 Proof of Proposition 1

F.2 Qubitisation: Chebyshev polynomials of the walk

F.3 The controlled walk

F.4 The imaginary-time filter

$$e^{-\beta x} = I_0(\beta) + 2 \sum_{k=1}^{\infty} (-1)^k I_k(\beta) T_k(x). \quad (\text{F.1})$$

F.5 GQSP phases

F.6 The readout identity for probabilistic preparation

G The Descriptor Construction in Full

G.1 Block rows in full

G.2 Higher order, variable coefficients, and the padded block count

For an order- k equation the state is $w = (\hat{f}^{(0)}; \dots; \hat{f}^{(k)})$ and A_{desc} stacks k recurrence block-rows $-\tilde{\mathbb{D}} \hat{f}^{(j)} + \tilde{\mathbb{B}} \hat{f}^{(j+1)} = 0$ above the single differential row $\sum_j \mathbb{M}_{a_j} \hat{f}^{(j)} = 0$: $k+1$ block-rows, each N coefficients wide, giving the classical $(k+1)N \times (k+1)N$ square matrix of Definition 11.

A variable coefficient $a_j(x)$ enters through $\mathbb{M}_{a_j} = \sum_q c_q \mathbb{M}_{x^q}$, $0 \leq q \leq N$, where c_q are the *monomial* coefficients of a_j : supplied directly as an array $[c_0, c_1, \dots]$, or, when a_j is given as a function, obtained by a least-squares fit to degree $N-1$ in the Chebyshev basis followed by an exact Chebyshev-to-monomial change of basis. The Chebyshev fit is what pins q to the same resolution N already fixed for the equation; its degree, not the subsequent change of basis, is the only place a_j 's representation is truncated. $\mathbb{M}_{x^q}$ is the resolution- n , square multiplication-by- x^q operator built from $x T_\ell = \frac{1}{2}(T_{\ell+1} + T_{\ell-1})$ with the T_N term dropped. $\mathbb{M}_{x^q}$ is banded with bandwidth exactly q , so it folds into the same shift-and-diagonal form as $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$. The dropped T_N term is the only approximation in the linear construction: for a smooth product resolved below degree $N-q$ the resulting error is confined to the top $\sim q$ Chebyshev modes and measures $\sim 10^{-13}$.

Padding to a power of two. The classical $(k+1)N \times (k+1)N$ matrix above is exact and complete as a linear system; it is not, however, the object any later stage of the pipeline actually builds. A gate-level realisation must index which block $\hat{f}^{(j)}$ a given operator acts on with a qubit register, and a register of m qubits addresses 2^m blocks, not $k+1$ of them for an arbitrary k . A_{desc} is therefore *embedded* in a $2^{\lceil \log_2(k+1) \rceil} N$ -dimensional space: the $k+1$ block-rows and

block-columns above are kept exactly as derived, and $2^{\lceil \log_2(k+1) \rceil} - (k+1)$ further diagonal blocks are appended, each an identity I_N pinning its own dummy field to zero, $\hat{f}_{\text{dummy}}^{(j)} = 0$. This padding is exact: the physical solution already satisfies the dummy rows trivially, since it has nothing in those blocks, so no error is introduced beyond the $\mathbb{M}_{x^q}$ truncation above. The qubit cost, however, is not uniform in k :

Order k	Blocks $k+1$	Field qubits $\lceil \log_2(k+1) \rceil$	Padded blocks $2^{\lceil \log_2(k+1) \rceil}$	Dummy blocks
1	2	1	2	0
2	3	2	4	1
3	4	2	4	0
4	5	3	8	3
7	8	3	8	0

so the state w the gate-level construction actually carries has $2^{\lceil \log_2(k+1) \rceil} N$ components, not $(k+1)N$: equal to the unpadded count exactly when $k+1$ is itself a power of two (orders $1, 3, 7, \dots$), and up to nearly double it otherwise – order 2, the most common case after order 1, already pays for a field register it only three-quarters uses. This is the qubit cost that must always accompany the gate-count comparison of Section 6.7.

G.3 Proof of Theorem 1

G.4 The term IR and its compilation

G.5 Proof of Proposition 5

G.6 Block rescaling

H The Nonlinear Route in Full

H.1 The doubled space

H.2 Proof of Proposition 6

H.3 The Carleman lift

H.4 Ratio rows and the space-time forms

H.5 Product folds

H.6 Fourier–Galerkin operators

I Verification Protocol

I.1 Modes, sizes and tolerances

I.2 Structured probes

I.3 Per-atom residuals

I.4 The classical reference: `eigh` against SVD

Table 15: Accuracy of the classical reference, $1 - |\cos|$ against the exact coefficients, for the standard form by `eigh`($R^\top R$) and by SVD of R , and for the descriptor form.

n	Standard, <code>eigh</code>	Standard, SVD	Descriptor
2	1.7×10^{-1}	1.7×10^{-1}	1.6×10^{-2}
3	8.6×10^{-8}	8.6×10^{-8}	5.8×10^{-9}
4	0	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
5	3.7×10^{-15}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
6	1.7×10^{-10}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
7	1.2×10^{-4}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
8	0.73	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$

I.5 The circuit-simulation memory and time model

I.6 Acceptance and the error budget

J Benchmark Catalogue

J.1 Linear problems

Table 16: The linear benchmark problems, R1–R5, with the paper’s printed invariant constraints.

	Equation	Invariant (printed)	Reference
R1	$f' + 2f = 0$	$-(f(-1) = 1 \text{ regular})$	$e^{-2(x+1)}$
R2a	$f'' + 4f' + 4f = 0$	$f(-1) = 0$	$\frac{1}{2}(1+x)e^{-2x}$
R2b	$f'' - 2f' - 3f = 0$	$f(0) = 0$	$\frac{1}{4}(e^{3x} - e^{-x})$
R2c	$f'' + 5f' + 400f = 0$	$f'(0) = 0$	$e^{-5x/2}(\cos \omega x + \frac{\sqrt{7}}{21} \sin \omega x),$ $\omega = \frac{15\sqrt{7}}{2}$
R3a	Legendre, $l = 4$	$f'(0) = 0$	P_4
R3b	Legendre, $l = 5$	$f(0) = 0$	P_5
R3c	associated Legendre, $l = 5, m = 1$	$f'(0) = 0$	P_5^1
R3d	associated Legendre, $l = 6, m = 1$	$f(0) = 0$	P_6^1
R4a	$(x-1)f'' - xf' + f = (x-1)^2$	$f'(-0.2) = 0$	closed form
R4b	$f'' + 4f' + 4f = e^{-2x}, \bar{p} = 3$	$f(-0.73) = 0$	closed form
R4c	$f'' - 5f' + 6f = xe^{2x}, \bar{p} = 5$	$f(0.91) = 0$	closed form
R4d	$f'' + 4f' + 20f = xe^{-2x} \sin 4x,$ $\bar{p} = 7$	$f'(-0.75) = 0$	closed form
R5a	Laplace, $f_{xx} + f_{yy} = 0$	$f(\pm 1, y) = f(x, -1) = 0$	$\cos \frac{\pi x}{2} \sinh \frac{\pi(y+1)}{2} / \sinh \pi$
R5b	heat, $f_t = \frac{1}{25}f_{xx}$	$f(t, \pm 1) = 0, f_x(t, \pm \frac{3}{4}) = 0$	$e^{-4\pi^2 t/25} \sin 2\pi x$
R5c	wave, $f_{tt} = 4f_{xx}$	$f(\frac{1}{8}, x) = f(\frac{7}{8}, x) = 0,$ $f(t, \pm 1) = 0$	$\cos 4\pi t \sin 2\pi x$

Table 17: The nonlinear and coupled benchmark problems, R6–R10.

	Equation	Data	Reference
R6a	$4f''' + 2f'^2 + f = 0$ (paper Eq. 29)	$f'(0) = 0, x_s = 0$	$1 - x^2/8$
R6b	$f'' - 2f^2 + x = 0$ (paper Eq. 34)	$f(0.02615) = 0$	boundary-value solve to 10^{-12}
R7	$u' = v, v' = -\omega^2 u, \omega = \pi$	$u(-1) = 0, v(-1) = \omega$	$u = \sin \omega(x + 1)$
R8	$y' = -y + y^2, t \in [0, 3]$	$y(0) = \frac{1}{2} (\rho = 0.5)$	$1/(1 + e^t)$
R9	viscous Burgers, periodic, $u_0 = a \sin x$	Fourier–Galerkin, $ k \leq K_c$	Cole–Hopf; DNS; exact lift
R10a	2-D vorticity Navier–Stokes, $\omega_0 = 2 \sin x \sin y$	$J \equiv 0$ (Taylor–Green)	exact viscous decay
R10b	2-D vorticity Navier–Stokes, vortex pair and resonant triads	active nonlinearity	DNS; exact lift

Table 18: State preparation by problem and form. Tier as marked.

	n	Infidelity (std)	Infidelity (desc)	Degree (std)	Degree (desc)	Tier
R1	—	—	—	—	—	
R2a	—	—	—	—	—	
R2b	—	—	—	—	—	
R2c	—	—	—	—	—	
R3	—	—	—	—	—	
R4	—	—	—	—	—	
R5	—	—	—	—	—	
R7	—	—	—	—	—	
R8	—	—	—	—	—	
R9	—	—	—	—	—	
R10	—	—	—	—	—	

Table 19: The paper’s solutions at its printed resolution: L^∞ error of the paper-faithful and corrected variants, and the corrected variant’s best over n . Classical computation.

Fig.	Problem	Printed n	Faithful	Corrected	Corrected, best over n
3b	R2a	3	$1.2 \times 10^{-3\dagger}$	$1.2 \times 10^{-3\dagger}$	3.1×10^{-13} ($n = 5$)
3c	R2b	3	$1.8 \times 10^{-2\dagger}$	$1.8 \times 10^{-2\dagger}$	1.8×10^{-11} ($n = 5$)
3d	R2c	5	2.2×10^{-3}	2.2×10^{-3}	2.1×10^{-11} ($n = 6$)
4a	R3a	3	1.3×10^{-15}	1.3×10^{-15}	1.3×10^{-15} ($n = 3$)
4b	R3b	3	1.6×10^{-15}	1.6×10^{-15}	1.6×10^{-15} ($n = 3$)
4c	R3c	5	$8.7 \times 10^{-1\dagger}$	1.7×10^{-14}	6.2×10^{-15} ($n = 4$)
4d	R3d	5	$1.2^\dagger$	1.4×10^{-13}	1.6×10^{-14} ($n = 3$)
5a	R4a	3	3.1×10^{-3}	1.8×10^{-6}	2.5×10^{-14} ($n = 4$)
5b	R4b	3	$1.9 \times 10^{-2\dagger}$	$7.8 \times 10^{-3\dagger}$	4.0×10^{-12} ($n = 5$)
5c	R4c	4	7.2×10^{-3}	4.9×10^{-10}	1.1×10^{-13} ($n = 5$)
5d	R4d	4	6.1×10^{-2}	4.9×10^{-7}	2.1×10^{-12} ($n = 5$)
6b	R5a	3	$3.9 \times 10^{-3\dagger}$	1.4×10^{-4}	1.9×10^{-13} ($n = 4$)
6d	R5b	4	1.9×10^{-4}	4.7×10^{-5}	2.8×10^{-12} ($n = 5$)
6f	R5c	5	8.5×10^{-8}	7.3×10^{-8}	1.4×10^{-10} ($n = 6$)
7b	R6a	2	$2.2 \times 10^{-16\dagger}$	$2.2 \times 10^{-16\dagger}$	—
7c	R6b	2	$6.8 \times 10^{-3\dagger}$	$6.8 \times 10^{-3\dagger}$	—

Table 20: Which printed choice causes the Fig. 5 error: the paper-faithful variant over $n = 4, \dots, 10$, each correction alone, and both together.

Fig.	Problem	Faithful	Exact root	Chebyshev source	Both, best
5a	R4a	3.1×10^{-3}	–	–	2.5×10^{-14} ($n = 4$)
5b	R4b	1.6×10^{-2}	4.0×10^{-2}	3.6×10^{-2}	4.0×10^{-12} ($n = 5$)
5c	R4c	7.2×10^{-3}	4.8×10^{-3}	1.0×10^{-2}	1.1×10^{-13} ($n = 5$)
5d	R4d	6.1×10^{-2}	8.1×10^{-2}	1.1×10^{-1}	2.1×10^{-12} ($n = 5$)

Table 21: The claims of Wu et al. tested in this thesis, and their verdicts.

	Claim	Verdict
A-01	$\mathbb{G}$, $\mathbb{M}$, $\mathfrak{n}_1$ and $\mathbb{B}$ as printed at $n = 2$	reproduced
A-02	the printed η_e of Figs 3–5 (11 panels)	within one unit of the last printed digit; 9/11 equal when rounded
A-03	the printed η_e of Fig. 6	not reproduced
A-04	the printed η_e of Fig. 7	not reproduced
A-05	the printed Laplace solution	a typographical error (cosh for cos)
A-06	the $\mathbb{B}$ circuit (Fig. 2): $n + 2$ qubits, prefactor $\sqrt{2^{n+1}}$	confirmed
A-07	the $\mathbb{G}$ circuit (Fig. 9): $2n + 3$ qubits, $O(n^3)$ gates, error decreasing exponentially	–
A-08	the $\mathbb{D}^{(0)}$ circuit (Fig. 8): prefactor $\sqrt{2^{n+1}}/f(x_s)$	confirmed
A-09	the linear H is positive definite with distinct eigenvalues	–
A-10	the doubled-space H has at least 2^{2n-1} zero eigenvalues	–
A-11	the time rule $t \geq \lambda_{\max}/(\lambda_2(n-1))$ suffices	–
A-12	the necessary terms scale as $O(\text{poly}(n))$ and are efficient	true for gates; not for the subnormalisation
A-13	the examples are solved accurately at the printed n	distorted where rounded points or Maclaurin sources enter

J.2 Nonlinear and coupled problems

J.3 Results by problem

K Reproduction Record

K.1 All panels

K.2 The Fig. 5 triage

K.3 Panels not reproduced, and printed errors

K.4 The paper's claims

K.5 The tau-truncation ablation

Table 22: Dropping the top k rows of the equation block (the Chebyshev tau method) gives an exact kernel with accuracy comparable to the paper's least squares.

Problem	n	Least-squares L^∞	σ_0	Tau-truncated L^∞	σ_0
R2a	5	3.1×10^{-13}	3.0×10^{-11}	5.7×10^{-13}	0
R2a	6	2.9×10^{-12}	4.9×10^{-10}	4.8×10^{-12}	0
R2a	7	3.0×10^{-11}	7.8×10^{-9}	1.1×10^{-10}	0
R2b	5	1.8×10^{-11}	5.3×10^{-11}	2.0×10^{-12}	0
R2b	6	8.3×10^{-11}	8.5×10^{-10}	3.8×10^{-11}	0
R2b	7	2.6×10^{-9}	1.4×10^{-8}	3.6×10^{-10}	0
R2c	5	2.2×10^{-3}	1.6×10^{-2}	1.8×10^{-3}	0
R2c	6	2.1×10^{-11}	5.5×10^{-10}	2.3×10^{-11}	0
R2c	7	6.3×10^{-10}	1.1×10^{-8}	4.2×10^{-11}	0

L Software and Reproducibility

L.1 Architecture

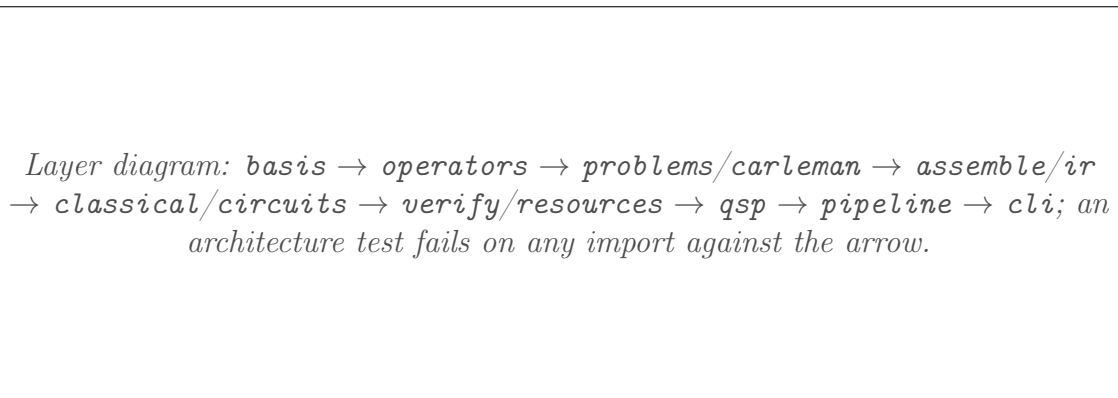


Figure 13: The layers of the `pihm` package and the one direction in which they may import.

L.2 Records and provenance

L.3 Campaigns and computational resources

Table 23: Resource classes for the Setonix campaigns.

Class	Request	Typical runs
S	work , 1 CPU, 4 GB, 30 min	construction, IR checks, small exact verification
M	work , 16 CPUs, 64 GB, 4 h	exact verification to 20 qubits; mid-size exact emulation
L	work , 64–128 CPUs, ~230 GB, 12–24 h	circuit-simulation ceilings; large emulation; probes to 30 qubits
XL	highmem , 1 TB	the largest emulation and probe runs

L.4 Tests

Table 24: The test tiers of the `pihm` package.

Tier	Content	Run
Unit and property	operator identities over random n and x ; primitives; arithmetic; IR; filters; phases	every commit
Golden	fixtures from the frozen legacy code and the paper’s printed $n = 2$ matrices	every commit
Reproduction	the paper’s targets, both variants	every commit
Integration	every problem, both forms, smallest n , all stages	every commit
Documentation	Sphinx build, doctests, spelling	every commit
Claims	one test per claim; writes the claims report	at gates
Slow	larger- n verification; circuit-simulation ceilings	at gates, on Setonix

L.5 Regenerating the results

L.6 Environment

M Research Proposal

Masters Research Proposal

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The University of Western Australia

May 25, 2025

Title: Solving Nonlinear PDEs via Quantum Singular Value Transformation

Supervisor: Professor Jingbo Wang

1 Introduction

1.1 Aims

This thesis aims to develop and evaluate quantum algorithms for solving nonlinear partial differential equations (PDEs) using the Quantum Singular Value Transformation (QSVT) framework. Based on recent work that represents differential equations and their boundary conditions as effective Hamiltonians, this work extends the QSVT to the Generalised Quantum Singular Value Transformation (GQSVT). This extension enables the application of this method to complex-valued differential equations which arise naturally in many areas of physics and engineering. Additionally the project will explore how to incorporate data-informed modifications into these Hamiltonians, with the goal of modelling data-rich regimes.

1.2 Significance

Solving complex nonlinear partial differential equations (PDEs) on quantum hardware is a rapidly emerging prospect with broad applications ranging from fluid dynamics and turbulence modelling to quantum field theory. Established algorithms such as the Harrow-Hassidim-Lloyd (HHL) algorithm and variational methods typically rely on discretisation, which inhibits their scalability to stiff, nonlinear, multiscale or dense systems with fine grid sizes. In contrast, the effective Hamiltonian approach uses spectral methods to encode the solution into a Chebyshev polynomial basis, allowing it to surpass performance when fine-grid encoding is required. The quantum singular value transformation (QSVT) provides a flexible and modular approach to quantum linear algebra, allowing operations like matrix inversion under broader conditions than HHL [1]. However, the current method restricts effective Hamiltonians to real values, thereby limiting the class of operators and solutions to purely real outcomes. Moreover, the HHL paper only addresses linear partial differential equations or nonlinear ordinary differential equations

through a process called linearisation [3] and variational algorithms require classical optimisation [7].

This research is significant because it extends the effective Hamiltonian based framework to complex-valued nonlinear PDEs, which are common in frontier areas of various fields. In domains like geophysical inversion, deep turbulence and real-time weather prediction, incorporating data-informed terms into the Hamiltonian provides a promising path towards optimising quantum solutions to problems where complete physical models are unavailable or difficult to create. By enabling hybrid physics and data driven Hamiltonians, this work could expand the applicability of quantum PDE solvers to data-rich, real world problems.

2 Background

2.1 Problem Statement

Many physical systems are described by partial differential equations (PDEs), but solving them remains computationally costly. In the absence of closed-form solutions, many advanced classical algorithms implement numerical methods which are difficult to scale and lose accuracy in systems with a need for more fine grid or time-step discretisations. This is especially true of nonlinear PDEs which are the primary focus of this research.

Quantum algorithms provide a potentially scalable alternative. Quantum Singular Value Transformation (QSVT) allows quantum circuits to encode real-valued polynomials of definite parity. Recent theoretical developments have shown that it is possible to represent differential equations and their boundary conditions as effective Hamiltonians, which are structured matrices that can be processed using QSVT. However, due to QSVT's restriction to real-valued Hermitian Hamiltonians, the range of solvable PDEs is restricted. Systems with damping, gain/loss or complex boundary interactions can lead to complex-valued or non-Hermitian operators.

While novel frameworks such as the Generalised QSVT (GQSVT) have been proposed to handle complex-valued operators and remove parity constraints, they have not been systematically applied to nonlinear PDEs.

The existing effective Hamiltonian approach is largely physics-driven, assuming known governing models. Yet in many high-dimensional, data-rich domains such as geophysical imaging, real-time weather prediction, or turbulence modelling, the governing equations may be incomplete or unavailable. In these settings, data-informed Hamiltonian construction offers a promising path forward towards more flexible and accurate quantum solvers.

This research addresses the gap in current quantum PDE solving frameworks by developing and applying GQSVT based methods for nonlinear, complex-valued PDEs and exploring hybrid physics and data driven constructions for real world data-intensive applications.

2.2 Literature Review

2.2.1 Exploration of Problems

Burgers Equation (Viscous)

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2} \quad (1)$$

This equation appears in turbulence theory and has a closed form solution in one real spatial dimension with $\nu > 0$. However, complex versions do exist and the equation remains unsolvable analytically outside these circumstances. The nonlinearity term is $u \frac{\partial u}{\partial x}$.

Nonlinear Schrödinger Equation

$$i \frac{\partial \psi}{\partial t} + \nabla^2 \psi + |\psi|^2 \psi = 0 \quad (2)$$

This equation describes the evolution of complex wave functions in nonlinear media. Its nonlinearity term is $|\psi|^2 \psi$.

Complex Ginzburg-Landau Equation

$$\frac{\partial A}{\partial t} = A + (1 + i\alpha) \nabla^2 A - (1 + i\beta) |A|^2 A \quad (3)$$

This equation models phenomena such as superconductivity and fluid convection patterns. The nonlinearity term is $|A|^2 A$.

The current physics-informed Hamiltonian method is able to address Burgers equation in its real case, but the Schrödinger and Ginzburg-Landau equations involve are complex-valued fields. This limitation motivates the development of a more robust framework capable of handling a broader class of nonlinear PDEs.

2.2.2 Existing Classical and Quantum Solving Methods

Classical methods for solving PDEs employ mesh based discretisation techniques such as finite difference and finite element methods or spectral methods [11]. These methods approximate the continuous domain onto a discrete grid, converting the PDE into a large system of algebraic equations that can be solved numerically. These methods are well-established and reliable, but for high-resolution or multi-dimensional systems the computational cost increases rapidly. For higher dimensional problems, direct matrix or classical variational methods may be employed [4].

The Harrow-Hassidim-Lloyd (HHL) algorithm was one of the first to demonstrate exponential speedup for solving systems of linear equations, which arise from discretised PDEs [3]. HHL encodes the linear system $A\vec{x} = \vec{b}$ into a quantum state and uses phase estimation to prepare a quantum state proportional to the solution vector $\vec{x}$. However, it is limited to sparse, well-conditioned Hermitian matrices. This makes it impractical for solving PDE problems, particularly nonlinear ones. Importantly, in the 2021 "Grand Unification" paper, it is shown that HHL can be viewed as a special case of QSVT, where

we're looking to find a target polynomial that inverts the singular values of the matrix A .

Variational quantum algorithms use a hybrid quantum-classical loop to optimise a parameterised quantum circuit that approximates the solution to a PDE [7]. This approach also discretises the domains into a grid. While more flexible than HHL, these methods require classical optimisation and suffer from challenges such as barren plateaus and scaling issues for larger systems.

2.2.3 Quantum Singular Value Transformation

The quantum singular value transformation (QSVT) is a powerful framework that generalises quantum linear algebra operations by enabling quantum circuits to apply polynomial transformations to the singular values of block encoded matrices [1]. QSVT enables the application of any parity-constrained polynomial $f(\Sigma)$ to the singular values Σ of a matrix A block encoded within a unitary.

$$U = \begin{bmatrix} A & \cdot \\ \cdot & \cdot \end{bmatrix} \text{ with } \|A\| \leq 1 \quad (4)$$

Assuming we have access to such a unitary U and the ability to apply a sequence of controlled-phase rotations by an angle ϕ , we can construct a unitary $V_{\vec{\phi}}$, dependent on the values of $\vec{\phi}$ we select for our phase rotations, that looks like the following:

$$V_{\vec{\phi}} = \begin{cases} \tilde{\Pi}_{\phi_1} U \left[\prod_{k=1}^{(d-1)/2} \Pi_{\phi_{2k}} U^\dagger \tilde{\Pi}_{\phi_{2k+1}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for odd } \text{Poly}^{(\text{SV})}(A) \\ \left[\prod_{k=1}^{d/2} \Pi_{\phi_{2k-1}} U^\dagger \tilde{\Pi}_{\phi_{2k}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for even } \text{Poly}^{(\text{SV})}(A) \end{cases} \quad (5)$$

where the sequences of phases $\vec{\phi} = (\phi_0, \phi_1, \dots, \phi_d)$ is chosen specifically to encode a desired polynomial $f(x)$ of degree d [1]. $\tilde{\Pi}_\phi$ and Π_ϕ represent a controlled phase rotation by ϕ where $\tilde{\Pi}$ and Π are the left and right unitary projectors respectively such that $A := \tilde{\Pi} U \Pi$. By carefully selecting which phase angles to rotate by, we can approximate any real, fixed parity continuous polynomial [6] acting on the singular values of A .

It can be shown that QSVT serves as a unifying framework for many quantum algorithms including quantum search, Hamiltonian simulation, factoring, quantum walks and more [8]. A key limitation of QSVT is that it only supports real-valued polynomials applied to matrices. This restricts its applicability to problems where both the matrix and the function being applied are real-valued [1]. For differential equations, this implies only PDEs with real-valued operators and real-valued solutions can be handled.

For physics-informed partial differential equation (PDE) solvers, QSVT is used in ground-state preparation via Hamiltonian simulation. In this case the target polynomial is $p(x) = e^{-tx}$ [11].

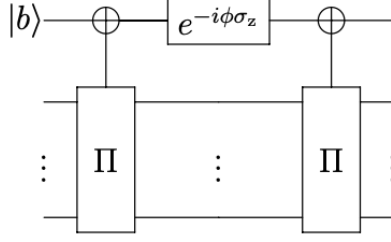


Figure 1: Quantum Circuit Diagram of the implementation of a Projector Controlled Phase Gate, Π_ϕ [2]

2.2.4 Physics-Informed Effective Hamiltonians

This algorithm encodes a differential equation in a latent Chebyshev basis and finds its solution by preparing the ground state of a specifically constructed Hamiltonian [11]. First, consider some differential equation with a differential operator $\mathcal{L}$, solution $u(x)$ and source function $f(x)$:

$$\mathcal{L}[u(x)] = f(x), \quad x \in [a, b]. \quad (6)$$

We can represent the solution as a state overlap in the Chebyshev basis with a scale factor s , namely

$$u(x) = s \langle T(x) | \psi \rangle \quad (7)$$

Here $|T(x)\rangle$ is known as a "feature map" state, a superposition of computational basis states with Chebyshev coefficients. $|\psi\rangle$ is a parameterised quantum state carrying the unknown amplitudes. Thus, we've effectively represented our solution in a latent Chebyshev basis [11].

To translate differential operators, we use the known differentiation matrices associated with the Chebyshev basis [5]. An example translation is shown below where $\mathbb{G}$ is the generator matrix corresponding to differentiation:

$$\frac{d^2 f(x)}{dx^2} + a f(x) = 0 \longrightarrow \mathbb{A} = \mathbb{G}^T + a \mathbb{I}. \quad (8)$$

Here, $\mathbb{A}$ is the matrix corresponding to the latent space representation of the DE. Crucially, differential operators act simply in the Chebyshev basis, as the following

$$\frac{d}{dx} f(x) = \frac{d}{dx} \langle T(x) | \psi \rangle = \langle T(x) | D^\top | \psi \rangle. \quad (9)$$

As a result, all terms in the DE can be rewritten as linear operators acting on $|\psi\rangle$.

To implement constraints, we use $\mathbb{B}(x)$, a data transformation operator dependent on x . For example, for specific constraints such as $f(x_0) = 0$ or $f'(x_1) = 0$, we can use $\mathbb{B}(x_0)$ or $\mathbb{B}(x_1)\mathbb{G}^\top$ [11]. From these, we can create our effective Hamiltonian:

$$\mathbb{H} = \mathbb{A}^\top \mathbb{A} + \mathbb{B}^\top \mathbb{B}. \quad (10)$$

We may add additional boundary constraints as required, which would effectively be more $\mathbb{B}$ matrices. Therefore, by construction, solving the original DE is equivalent to solving the eigen-problem:

$$\mathbb{H}_{\text{eff}}|\psi\rangle = 0, \quad (11)$$

and the desired solution corresponds to the ground state of the effective Hamiltonian. $\mathbb{H}$ is a real symmetric positive-definite matrix with distinct eigenvalues and mutually orthonormal eigenvectors [11].

We then block encode the effective Hamiltonian, and use a ground state preparation routine, QSVT, to amplify the amplitude of the ground state. Specifically, we use quantum imaginary time evolution, constructing the polynomial approximation of $e^{-t\mathbb{H}}$. This exponentially suppresses the excited eigen-components, leaving a higher probability of measuring the ground state. Importantly, the algorithm directly prepares the lowest energy solution without classical optimisation [11].

Finally, to measure the quantum state, we use a Chebyshev transformation and Hadamard test to measure the state we have prepared. This controlled overlap test effectively measures the overlap $\langle T(x)|\psi\rangle$. In practice this is done via interferometric measurement [11].

This same framework is extendable to multi-variable and nonlinear DEs. For a function of two variables $f(x, y)$, we use a tensor product Chebyshev basis with two registers, preparing a joint feature state $|T(x)\rangle \otimes |T(y)\rangle$. Take Laplace’s equation for example:

$$\frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = 0 \longrightarrow \mathbb{A} = \mathbb{G}_x^{\top 2} \otimes \mathbb{G}_y^{\top 2}. \quad (12)$$

For nonlinear terms, we add extra registers. For example, a term like $f'(x)^2$ leads to a product of two copies of $|\psi\rangle$ in latent space. We create a new, higher dimensional latent Chebyshev space where $|\Psi\rangle \equiv |\psi\rangle \otimes |\psi\rangle$, so that the nonlinearity term becomes linear in the larger space. This naturally handles all nonlinearity. Inhomogeneous terms are handled via Maclaurin series expansion. For all these cases, we then construct the effective Hamiltonian as usual and solve [11]. This approach allows us to encode the solution over the entire domain, rather than using grid discretisation. Therefore, it is possible to evaluate the solution function at any arbitrary input values. However, as discussed above, QSVT can only handle real values. This restricts the class of differential equations we’re able to solve.

Example Solutions:

Below are results from the application of this algorithm on example differential equations [11]. The paper did not attempt to solve nonlinear PDEs, but there are examples of both linear PDEs and nonlinear DEs. $f(x)$ is the known solution and $f_Q^*(x)$ is the quantum solution.

1. Heat Equation

$$\frac{\partial f(t, x)}{\partial t} - k \frac{\partial^2 f(t, x)}{\partial x^2} = 0, k = \frac{1}{25}, \mathcal{BC} = \{f(0, x) = \sin(2\pi x), f(t, \pm 1) = 0\} \quad (13)$$



Figure 2: Solution for the Heat Equation Example. [11]

2. Nonlinear Example 1

$$4 \frac{d^2 f(x)}{dx^2} + 2 \left(\frac{df(x)}{dx} \right)^2 + f(x) = 0, \mathcal{BC} = \{f(0) = 1, f'(0) = 0\} \quad (14)$$

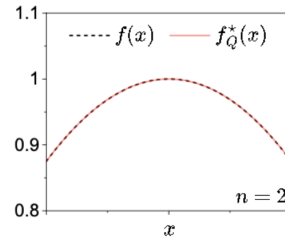


Figure 3: Solution of Nonlinear Example 1 [11]

3. Nonlinear Example 2

$$\frac{d^2 f(x)}{dx^2} - 2f(x)^2 + x = 0, \mathcal{BC} = \{f(-1) = -0.1, f(1) = 0.1\} \quad (15)$$

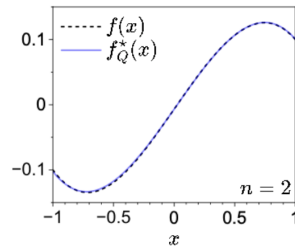


Figure 4: Solution of Nonlinear Example 2 [11]

The model performs well for all examples shown. The errors are almost negligible and decrease as the number of qubits used (shown as n on the figures) increase. The experiments were conducted on classical simulations of quantum hardware and are therefore currently computationally costly, however based on these results it is projected this algorithm will become useful in the early fault-tolerant era [11].

2.2.5 Generalised Quantum Singular Value Transformation

The generalised QSVT (GQSVT) framework extends the standard QSVT algorithm by lifting the restrictions on coefficient phases and polynomial parity [9]. The standard QSVT restricts the target polynomial $p(x)$ to have real coefficients and definite parity. However, the generalised version allows complex polynomials and indefinite parity. These generalisations are inherited from developments in generalised quantum signal processing (GQSP) which is itself an extension of the quantum signal processing framework that QSVT was built on [10]. Eliminating the restrictions on real polynomials and Hamiltonians imposed by QSVT will allow us to solve complex-valued differential equations. GQSVT also promises a simplification in complexity for the crucial and time-consuming step of calculating phase factors. The task was of complexity $\mathcal{O}(d^2)$ for QSVT, GQSVT promises a complexity of $\mathcal{O}(d)$ [9]. This should speed up calculations even for problems which are currently possible via the existing method.

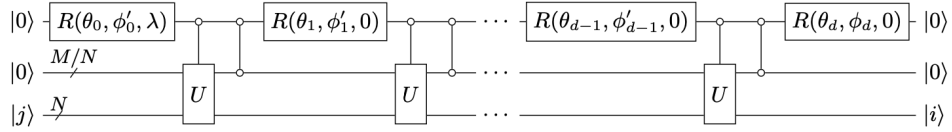


Figure 5: Example circuit for the generalised quantum singular value transformation for a Hermitian matrix A , block encoded as U . R is a rotation matrix analogous to Π for QSVT [9].

3 Research Plan

3.1 Method

This research will be primarily theoretical and computational. The goal is to design, simulate and evaluate novel quantum algorithms for solving nonlinear, complex-valued partial differential equations (PDEs). Development will be conducted using *Mathematica* for symbolic manipulation and proof-of-concept derivations. Simulations will be created using *Python* based simulators such as *Qiskit* and *PennyLane* and results will be compared to classical solving algorithms. Efforts will be split into two phases. First, extending the current QSVT algorithm to GQSVT and second, integrating data-informed terms into effective Hamiltonians to enable solving PDEs in data-rich regimes. This research represents a new research direction within QUISA.

Proposed Plan:

1. Reproduce the Current Physics-Informed QSVT Method:
 - (a) Implement the standard QSVT framework in *Python*.
 - (b) Choose an appropriate problem.
 - (c) Implement a physics-informed Hamiltonian PDE solver in *Python*.
 - (d) Compare quantum solution against classical benchmarks
2. Extend to GQSVT:
 - (a) Create the GQSVT algorithm in *Python*.
 - (b) Choose an appropriate problem.
 - (c) Implement a new physics-informed Hamiltonian PDE solver in *Python*.
 - (d) Compare quantum solution against classical benchmarks
3. Experiment with more data-informed terms in the Hamiltonian

Timeline:

Semester 1 2025: Research Proposal.

Semester 2 2025: Recreation of QSVT Physics-informed method. Begin extension to GQSVT.

Semester 1 2026: Completion of a working GQSVT enhanced algorithm, including simulation and evaluation in comparison to classical algorithms.

Semester 2 2026: Experimentation with data-informed terms.

3.2 Preliminary Work

In semester 1 2025, work consisted of a thorough literature review, achieving an in-depth understanding of QSVT and the physics-informed Hamiltonian approach for PDEs. This was pivotal in developing a thesis that could both build on the newest developments in the field and be achievable within the time-frame of a Master's course. A working *Python* implementation of QSVT has been created and work is underway on implementing the current physics-informed Hamiltonian approach using QSVT.

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Table 25: Deviations from the research proposal of May 2025.

Proposed	Outcome	Reason
Extending QSVT to the generalised construction for complex-valued, non- Hermitian operators Nonlinear PDEs: Burgers, nonlinear Schrödinger, com- plex Ginzburg–Landau Data-informed Hamiltonians (not proposed)		